Revisiting ASkewSGD: Its New Theoretical Guarantees for Quantization-Aware Deep Neural Network Optimization

Hok Fong Wong
Department of Computer Science and Engineering, The Chinese University of Hong Kong

Hoi-To Wai
Department of Systems Engineering and Engineering Management, The Chinese University of Hong Kong

Abstract

The question of enforcing quantization on the weights and activations in Deep Neural Networks (DNNs) has come to the forefront in recent years due to its relevance to restricted memory and/or computational resources scenarios. While low-precision fixed integer values could substantially reduce the memory footprint and latency, inaccurate quantization on model parameters is susceptible to a significant accuracy drop. In this paper, we strive to improve our understanding of quantization-aware training algorithms and provide stronger guarantees for the Annealed Skewed Stochastic Gradient Descent algorithm (`ASkewSGD`) proposed by Leconte et al. (2023). In particular, we attempt to retain the algorithm's convergence guarantees without hinging on a highly differentiable loss function. We supplement the numerical experiments of Leconte et al.'s (2023) and justify that `ASkewSGD` is able to surpass or perform on par with the classical benchmarks, justifying its effectiveness as a robust optimization algorithm for DNN quantization.

Keywords: Analysis of Algorithms, Non-Convex Optimization, Quantization-Aware Training

1 INTRODUCTION

1.1 Towards the Time- and Energy-Efficient, Optimal Neural Network Design

Over the past decade, there has been an increasing awareness that deep learning models have succeeded in achieving vast improvements for tasks such as computer vision, speech recognition and generation, and natural language processing. While the accuracy of these models has significantly improved, the size of these models could be a potential problem when being deployed in resource-limited scenarios. As such, there is a growing interest in the search for efficient design, training, and deployment of neural network models, while allowing optimality trade-offs. Methods include knowledge distillation, micro-architecture design, hardware-architecture co-design, pruning of insensitive neurons, low-rank decomposition, and network quantization (Gholami et al., 2021). A recent paper also demonstrated the robustness of quantization over neuron pruning (Kuzmin et al., 2023). In this paper, we primarily focus on quantization solutions during model training, where weights are mapped from real values to a finite set of discrete values with lower bit widths, resulting in storage-friendly neural networks.

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1.2 Contributions

- We improve the understanding of `ASkewSGD` and provide new convergence insights for `ASkewSGD` proposed by Leconte et al. (2023) under weaker assumptions. We identify a loophole in their proof on the Lipschitzness condition and further weaken the differentiability requirement of the loss function.
- We evaluate the performance of `ASkewSGD` along with other quantization-aware training methods (`BinaryConnect`, `ProxQuant`) by numerical experiments on `TwoMoons`, `MNIST` and `CIFAR-10`. We made the related codes available at `https://github.com/SWongHF/Experiment-2025-Summer`.

1.3 Notations

We use $\|\cdot\|$ to denote the Euclidean norm for vectors. We denote the i-th component of a weight vector $\mathbf{w}$

as w_i or $[\cdot]_i$, and the superscript $\mathbf{w}^{(k)}$ to denote the k-th iterate of the optimization algorithm. We denote a sample gradient of ℓ at $\mathbf{w}$ as $\widehat{\nabla}\ell(\mathbf{w})$ so that $\mathbb{E}[\widehat{\nabla}\ell(\mathbf{w})] = \nabla\ell(\mathbf{w})$.

2 PROBLEM STATEMENT AND RELATED WORKS

2.1 Formalization of the Optimization Problem and Inherent Difficulty

We are interested in solving the optimization problem related to learning a quantized neural network (QNN),

$$\min_{\mathbf{w}\in\mathcal{Q}\subseteq\mathbb{R}^d} \ell(\mathbf{w}), \text{ where } \ell(\mathbf{w}) = \mathbb{E}_{(\mathbf{x},y)\sim p_{\text{data}}}[L(\mathbf{f}(\mathbf{x},\mathbf{w}),y)], \tag{$\mathcal{P}$}$$

where d is the number of network weight parameters, $\mathbf{f} : \mathbb{R}^e \times \mathbb{R}^d \to \mathbb{R}^m$ models a neural network with the outputs a vector in $\mathbb{R}^m$ associated with its network weights $\mathbf{w} \in \mathbb{R}^d$ and the training sample $\mathbf{x} \in \mathbb{R}^e$, $L : \mathbb{R}^m \to \mathbb{R}$ denotes the training loss, $\mathcal{Q}$ is a discrete, non-convex set of quantization levels (possibly coordinate-wise defined), and p_{data} is the training distribution.

We do not necessarily aim to solve the above (hard, combinational) optimization problem ($\mathcal{P}$) globally and optimally. Instead, we want to find discrete weights $\mathbf{w} \in \mathcal{Q}$ that remain satisfactory when compared to the non-quantized continuous weights. This circumvents the NP-Hard complexity of MINLP (Kannan & Monma, 1978) as the structure of ℓ could come into aid: if ℓ is reasonably smooth, a closed-by quantized weight of a locally optimal continuous weight will likely yield similar performance (Gholami et al., 2021).

2.2 The BinaryConnect Algorithm

Courbariaux et al. (2015) considered an aggresive quantization scheme using binary networks, where $\mathcal{Q} = \{\pm 1\}^d$. The BinaryConnect algorithm updates the weights by the following method:

$$\begin{aligned} &\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} - \gamma_k \widehat{\nabla}\ell(\mathsf{P}(\mathbf{w}^{(k)})), \\ &\text{where } [\mathsf{P}(\mathbf{w})]_i = \begin{cases} +1, & w_i \geq 0, \\ -1, & \text{otherwise.} \end{cases} \end{aligned} \tag{1}$$

While the above method is deterministic, another scheme quantizes the weights stochastically, which is given by:

$$\begin{aligned} &[\mathsf{P}(\mathbf{w})]_i = \begin{cases} +1, & \text{with probability } \sigma(w_i), \\ -1, & \text{with probability } 1-\sigma(w_i), \end{cases} \\ &\text{where } \sigma(x) = \text{clamp}\left(\frac{x+1}{2}, 0, 1\right). \end{aligned} \tag{2}$$

For the general quantized neural network, BinaryConnect can easily be adapted by adding a quantization step that maps a real number input $w \in [0,1]$ to a k-bit number output w_q (Zhou et al., 2016):

$$w_q = \text{round}((2^k - 1)w) - (2^{k-1} - 1). \tag{3}$$

Despite its empirical success, the practical convergence of BinaryConnect remains poorly understood. It is pointed out by Bai et al. (2019) that the algorithm can result in non-convergence if no fixed-point is available. Whereas, the behaviour of BC is analyzed by Dockhorn et al. (2021), stating that the updates of BC are formally the same as the dual averaging (DA) algorithm (as a non-convex counterpart).

2.3 The ProxQuant Algorithm

Bai et al. (2019) proposed the proximal gradient method for quantization-aware training, which is a variant of the proximal operator

$$\mathbf{w}^{(k+1)} \leftarrow \mathsf{P}\left(\mathbf{w}^{(k)} - \eta_k \widehat{\nabla}\ell(\mathbf{w}^{(k)})\right), \tag{4}$$

where the proximal operator is defined as

$$\begin{aligned} &\mathsf{P}(\mathbf{w}) = \operatorname*{arg\,min}_{\bar{\mathbf{w}}\in\mathbb{R}^n} \left(\frac{1}{2}\|\mathbf{w} - \bar{\mathbf{w}}\|^2 + \lambda R(\bar{\mathbf{w}})\right), \\ &\text{where } R(\bar{\mathbf{w}}) = \inf_{\widehat{\mathbf{w}}\in\mathcal{Q}} \|\widehat{\mathbf{w}} - \bar{\mathbf{w}}\|. \end{aligned} \tag{5}$$

This method is theoretically sound, but the proximal operator can be expensive to compute when $\mathcal{Q}$ is large.

3 The ASkewSGD Algorithm

3.1 Algorithm Description

Inspired by the success of ProxQuant, Leconte et al. (2023) considers a variant of the MJ algorithm (Muehlebach & Jordan, 2022), based on local approximation for the discrete constraints. They break down the original problem into a smoothed sequence of interval-constrained optimization problems $(\mathcal{P}_\varepsilon)_{\varepsilon>0}$.

In particular, they shed light on the following optimization problem (common for supervised learning):

$$\min_{\mathbf{w}\in C_\varepsilon} \ell(\mathbf{w}) := \frac{1}{N}\sum_{j=1}^{N} \ell_j(\mathbf{w}), \tag{$\mathcal{P}_\varepsilon$}$$

where $C_\varepsilon := \{\mathbf{w} \in \mathbb{R}^d \mid g_{\varepsilon,i}(\mathbf{w}) \geq 0, i = 1, 2, \ldots, c\}$.

Here, N is the size of the training set, $\ell_j : \mathbb{R}^d \to \mathbb{R}$ is the loss associated with the j-th observation, $g_{\varepsilon,i} : \mathbb{R}^d \to \mathbb{R}^n$ is the i-th constraint with dependence on ε, which is a given parameter controlling the landscape of $g_{\varepsilon,i}$.

Definition 1 (**Quadratic Remoteness Measure of Iterates to the Quantization Set**). *Let $\varepsilon \in [0,1]$, $w_i \in \mathcal{Q}_i, i \in [d]$, where $\mathcal{Q}_i = \{q_i^{(1)}, \ldots, q_i^{(K_i)}\}$ are sets of quantization values defined coordinate-wise. We define the piecewise function*

$$\psi_i(w_i;\varepsilon) := \begin{cases} \varepsilon - (q_i^{(1)} - w_i), & w_i < q_i^{(1)}, \\ \varepsilon - (q_i^{(j-1)} - w_i)(w_i - q_i^{(j)}), & \\ \quad q_i^{(j-1)} \leq w_i < q_i^{(j)}, j = 2, \ldots, K_i, & \\ \varepsilon - (w_i - q_i^{(K_i)}), & w_i \geq q_i^{(K_i)}, \end{cases} \tag{6}$$

for all $w_i \in \mathbb{R}$.

Note that Leconte et al. (2023) uses a quartic version. We particularly define ψ as a quadratic function instead, since our version is Lipschitz-smooth. The proof of Leconte et al. (2023) still applies. We illustrates the special cases with Figure 1 and Figure 2 above.

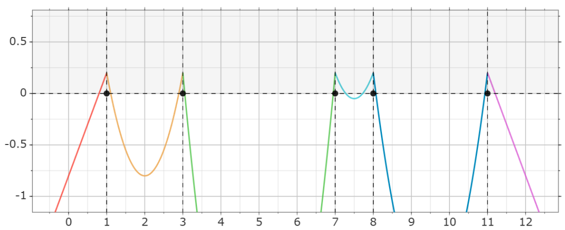

Figure 1: $\psi_i(w_i;\varepsilon)$, where $\varepsilon = 0.2$ and $\mathcal{Q}_i = \{1, 3, 7, 8, 11\}$.

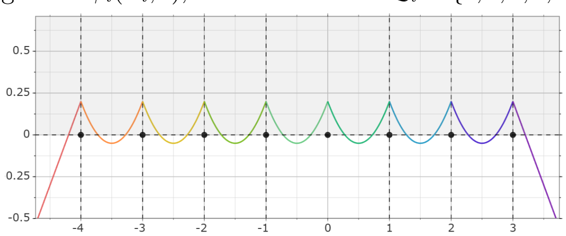

Figure 2: The INT3 uniform quantization scheme, where $\varepsilon = 0.2$ and $\mathcal{Q}_i = \{-4, -3, -2, -1, 0, 1, 2, 3\}$.

By setting c (the number of constraints) as d (the number of parameters with their respective quantization requirements) and $g_{\varepsilon,i}(\mathbf{w}) = \psi_i(w_i;\varepsilon)$, we obtain a compact set C_ε, which is also non-empty. Furthermore, as $\varepsilon \to 0^+$, the set C_ε recovers the original quantization set $\mathcal{Q}$. As a consequence, we can solve the QNN optimization problem by considering the smoothed sequence of interval-constrained optimization problems $(\mathcal{P}_{\varepsilon_j})_{j\in\mathbb{N}}$ with $\lim_{j\to\infty} \varepsilon_j = 0$, while each $(\mathcal{P}_{\varepsilon_j})$ is solvable by a variant of the MJ Algorithm.

Definition 2 (**MFCQ**). *We say that the Mangasarian-Fromovitz constraint qualification (MFCQ) holds, if and only if $\forall \mathbf{w} \in \mathbb{R}^d$, there exists $\mathbf{v} \in \mathbb{R}^d$, such that $\mathbf{v}^\top \nabla g_{\varepsilon,i}(\mathbf{w}) > 0$ for all $i \in I(\mathbf{w}) := \{i \in [c] \mid g_{\varepsilon,i}(\mathbf{w}) \leq 0\}$.*

As the constraint function $g_{\varepsilon,i}$ is coordinate-wise independent and we can always follow the gradient direction of $g_{\varepsilon,i}$ on the i-th coordinate whenever the constraint is activated, it is easy to verify that MFCQ always holds, unless there exists an index $i \in [d]$ such that $w_i = (q_i^{(j)} + q_i^{(j+1)})/2$. However, the set of such w is Lebesgue-measure zero, and the algorithm will also guarantee that we will never stumble upon such a point.

We then introduce some essential concepts from nonconvex optimization with respect to MFCQ.

Definition 3 (**Tangent and Normal Cones**). *The Clarke's tangent cone of C at $\mathbf{x}$ contains all $\boldsymbol{\delta}\mathbf{x} \in T(\mathbf{x})$ if there exists two sequences $\mathbf{x}_j \to \mathbf{x}, \mathbf{x}_j \in C, t_j \downarrow 0$ such that $(\mathbf{x}_j - \mathbf{x})/t_j \to \boldsymbol{\delta}\mathbf{x}$. The normal cone at $\mathbf{x}$ is subsequently defined as*

$$N(\mathbf{x}) = \{\boldsymbol{\lambda} \in \mathbb{R}^n \mid \boldsymbol{\lambda}^\top \boldsymbol{\delta}\mathbf{x} \leq 0, \forall \boldsymbol{\delta}\mathbf{x} \in T(\mathbf{x})\}. \tag{7}$$

Lemma 1. *Suppose that $\mathbf{x} \in C_\varepsilon$ and the Mangasarian-Fromovitz constraint qualification holds for every $\mathbf{x} \in C_\varepsilon$. Let T_{C_ε} be the Clarke's tangent cone of C_ε. Then, every $\boldsymbol{\delta}\mathbf{x} \in T_{C_\varepsilon}(\mathbf{x})$ should satisfy $\nabla g_{\varepsilon,i}(\mathbf{x})^\top \boldsymbol{\delta}\mathbf{x} \geq 0, \forall i \in I(\mathbf{x})$. The converse also holds.*

The proof is deferred to Appendix A. A direct consequence is the simplification of the tangent and normal cones of C_ε at $\mathbf{w} \in C_\varepsilon$, given by

$$T_M(\mathbf{w}) = \left\{\mathbf{v} \in \mathbb{R}^d : \nabla g_{\varepsilon,i}(\mathbf{w})^\top \mathbf{v} \geq 0, \forall i \in I(\mathbf{w})\right\}, \tag{8}$$

$$N_M(\mathbf{w}) = \left\{ -\sum_{i \in I(\mathbf{w})} \lambda_i \nabla g_{\varepsilon,i}(\mathbf{w}), \lambda_i \in \mathbb{R}_{++} \right\}. \tag{9}$$

Furthermore, since the MFCQ holds, the set of stationary points is given by the Karush-Kuhn-Tucker condition:

$$\mathcal{Z}_\varepsilon := \{\mathbf{w} \in C_\varepsilon : \mathbf{0} \in -\nabla \ell(\mathbf{w}) - N_{C_\varepsilon}(\mathbf{w})\}. \tag{10}$$

It then implies that $\mathbf{w} \in \mathcal{Z}_\varepsilon$ if and only if for all $i \in [d]$, $[\nabla \ell(\mathbf{w})]_i = 0$ when $\psi_i(w_i;\varepsilon) > 0$ and $\text{sign}([\nabla \ell(\mathbf{w})]_i) = \text{sign}(\psi_i'(w_i;\varepsilon))$ when $\psi_i(w_i;\varepsilon) = 0$.

Muehlebach and Jordan (2022) considers the approximation by an extention of the tangent cone "outside" of the feasible set at $\mathbf{w}$. For our problem, we have

$$V_{\varepsilon,\alpha}(\mathbf{w}) := \{\mathbf{v} \in \mathbb{R}^d : v_i \psi_i'(w_i;\varepsilon) \geq - \alpha \psi_i(w_i;\varepsilon) \text{ for } i \in I(\mathbf{w})\} \tag{11}$$

(*cf.* Equation (8)).

We are now ready to state the `ASkewSGD` algorithm. The basic idea is to "skew" the search direction in order to force the algorithm to find a minimizer without constraining the sequence $(\mathbf{w}^{(k)})_{k\in\mathbb{N}}$ to the feasible set by cleverly using the extended tangent cone $V_{\varepsilon,\alpha}(\mathbf{w})$, which reformulates the position constraints into forward "velocity" constraints.

Algorithm 1: `ASkewSGD` for QNN Training

for $k = 0, 1, \ldots$ **do**
- Obtain a stochastic gradient $\widehat{\nabla}\ell(\mathbf{w}^{(k)}) = 1/N_b \sum_{i=1}^{N_b} \nabla\ell_{j_i}(\mathbf{w}^{(k)})$.
- $\widehat{\mathbf{v}}^{(k)} \leftarrow \mathrm{clip}(\arg\min_{\mathbf{v}\in V_{\varepsilon,\alpha}(\mathbf{w}^k)}(1/2)\|\mathbf{v} + \widehat{\nabla}\ell(\mathbf{w}^{(k)})\|^2, M_c)$.
- $\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} + \gamma_k\widehat{\mathbf{v}}^{(k)}$.

In every iteration, the iterate $\mathbf{w}^{(k)}$ is updated by the velocity. The velocity $\mathbf{v}^{(k)}$ is chosen to match as closely to the gradient flow $-\nabla\ell(\mathbf{w}^{(k)})$ as possible, subject to the velocity constraint $\mathbf{v}^{(k)} \in V_{\varepsilon,\alpha}(\mathbf{w}^{(k)})$. This motivates us to find a skewed (stochastic) gradient $\widehat{\mathbf{v}}^{(k)}$ so that

$$\widehat{\mathbf{v}}^{(k)} = \underset{\mathbf{v}\in V_{\varepsilon,\alpha}(\mathbf{w}^{(k)})}{\arg\min} \frac{1}{2}\left\|\mathbf{v} + \widehat{\nabla}\ell(\mathbf{w}^{(k)})\right\|^2, \tag{12}$$

The above equation (12) admits an explicit solution:

$$[\mathbf{s}_{\varepsilon,\alpha}(\mathbf{g},\mathbf{w})]_i := \begin{cases} -g_i, & \text{if } \psi_i(w_i;\varepsilon) > 0 \text{ or} \\ \quad -g_i \cdot \psi_i'(w_i;\varepsilon) \geq -\alpha\psi_i(w_i;\varepsilon) \geq 0, & \\ -\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon), & \text{otherwise.} \end{cases} \tag{13}$$

To simplify notation, we denote $u_i := -\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon)$, which represents a constraint correction for parameter w_i. We refer to this as the *skewing force* hereafter. To ensure that $[\mathbf{s}_{\varepsilon,\alpha}(\mathbf{u},\mathbf{w})]_i$ does not diverge to infinity, we apply the clip operator to bound the update direction $\widehat{\mathbf{v}}^{(k)}$ within a fixed range controlled by M_c. We set $\widehat{v}_i^{(k)} = M_c$ if $w_i = (q_i^{(j)} + q_i^{(j+1)})/2$ by convention.

We note how the parameters $\mathbf{w}^{(k)}$ are allowed to escape from the feasible set temporarily in search of better local basins. This crucial fact has made the algorithm achieve better results when compared to the classical projected gradient descent algorithm.

3.2 Assumptions

We describe several classical assumptions made by Leconte et al. (2023) on the loss function ℓ and the step size sequence $(\gamma_k)_{k\geq 0}$ for the `ASkewSGD` Algorithm.

Assumption 1 (**Lipschitz Smoothness**). *For $j \in \{1,\ldots,N\}$, the function ℓ_j is continuously differentiable and M_{ℓ_j}-smooth, i.e.*

$$\|\nabla\ell_j(\mathbf{x}) - \nabla\ell_j(\mathbf{y})\| \leq M_{\ell_j}\|\mathbf{x} - \mathbf{y}\| \text{ for any } \mathbf{x},\mathbf{y} \in \mathbb{R}^d.$$

Additionally, we define $M_\ell := \sup_{1\leq j\leq N} M_{\ell_j}$.

Assumption 2 (**Lipschitzness**). *For $j \in \{1,\ldots,N\}$, the function ℓ_j is continuously differentiable and L_{ℓ_j}-Lipschitz continuous, i.e.*

$$|\ell_j(\mathbf{x}) - \ell_j(\mathbf{y})| \leq L_{\ell_j}\|\mathbf{x} - \mathbf{y}\| \text{ for any } \mathbf{x},\mathbf{y} \in \mathbb{R}^d.$$

Additionally, we define $L_\ell := \sup_{1\leq j\leq N} L_{\ell_j}$.

Assumption 3 (**Robbins-Monro Stepsizes**). *The stepsizes $(\gamma_k)_{k\geq 0}$ are positive, non-increasing, non-summable and square-summable, i.e. $\sum_{k=0}^\infty \gamma_k = +\infty$, and $\sum_{k=0}^\infty \gamma_k^2 < +\infty$.*

Assumption 4 (**Smooth Loss Landscape**). *For $j \in \{1,\ldots,N\}$, the function ℓ_j is d-times continuously differentiable.*

Remark. A2 can be restrictive as it rules out many functions (such as the quadratic function) as a consequence. A3 is a standard assumption in stochastic optimization, stating that non-summable and squared summable stepsizes are necessary for convergence. A4 is non-standard. It represents strong differentiability of the loss function's differentiability, which is not always the case for neural networks.

The optimization algorithm has a convergence guarantee summarized by the following theorem:

Theorem 1 (**Leconte et al., 2023**). *Assume that A1, A2, A3, A4 holds, and $0 < \varepsilon \leq \inf_{1\leq i\leq d}\inf_{1\leq j<K_i}|q_i^{(j)} - q_i^{(j+1)}|^2/4$, where $\{q_i^{(j)}\}$ are the quantization levels, $\ell(\mathbf{w}^{(k)})$ converges and $\lim_{k\to\infty} \mathrm{d}(\mathbf{w}^{(k)}, \mathcal{Z}_\varepsilon) = 0$ almost surely.*

Here, we have identified a loophole in Leconte et al.'s (2023) proof and corrected it with the addition of A2, due to the use of a bounded gradient property of the loss function. Leconte's proof relies heavily on a "tame" property as described by the subgradient method of Davis et al. (2020), where the Weak Sard's property of the loss function ℓ is fulfilled by assuming the high-order differentiability of the underlying loss function (A4).

Our work is to remove A2 and A4. We also attempt to investigate the general behavior of `ASkewSGD` without A3, since a fixed learning rate is often preferred in machine learning scenarios.

Leconte et al. (2023) also applied a technical lemma that describes the dynamics of the algorithm. We will make use of this fact, further improve it, and leave its proof in Appendix B.

Lemma 2. *Assume that A2, A3 holds, we have $\limsup_{k\to\infty} \mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) = 0$ almost surely.*

Lemma 2 states that the iterates will eventually converge to the feasible set. This is especially useful when we later show that the magnitude of the skewing force $\mathbf{u}^{(k)}$ can be bounded by the distance $\mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon)$. As we would expect, an iterate should not be aggressively pushed to the quantization set. Otherwise, it may either escape from the local loss basin or waste time on entering the feasible region. This gives a strong reason for us to include A3 in the lemma.

4 NEW CONVERGENCE GUARANTEES OF THE `ASkewSGD` ALGORITHM

In this section, we will provide new theoretical convergence guarantees for the `ASkewSGD` algorithm. First, we present our analysis on the deterministic case, assuming that the gradient oracle always returns an accurate result. Second, we leverage our proof on the deterministic case to the stochastic case with perturbed gradients under mild conditions.

To achieve the desired results, we aim to deepen our understanding of the algorithm's behavior. We prove that the true update direction $\mathbf{v} := \mathbf{s}_{\varepsilon,\alpha}(\nabla\ell(\mathbf{w}), \mathbf{w})$ can be a good indicator of optimality for $(\mathcal{P}_\varepsilon)$:

Lemma 3. *$\mathbf{w} \in \mathcal{Z}_\varepsilon$ if and only if $\mathbf{s}_{\varepsilon,\alpha}(\nabla\ell(\mathbf{w}), \mathbf{w}) = \mathbf{0}$.*

Now, we explain the choice of ψ_i (as defined in (6)) and establish its connection with the skewing force by the following lemma:

Lemma 4. *Fix an arbitrary $0 < \varepsilon \leq \inf_{1\leq i\leq d} \inf_{1\leq j<K_i} |q_i^{(j)} - q_i^{(j+1)}|^2/4$, where $\{q_i^{(j)}\}$ are the quantization levels. For $j \in \{2, \ldots, K_i - 1\}$, denote $[q_-, q_+]$ as the set $C_{\varepsilon,i} \cap [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$, where $C_{\varepsilon,i}$ is the projection of C_ε on the i-th coordinate. Let $0 < \delta_1 < (q_i^{(j)} + q_i^{(j+1)} - 2q_+)/3$ and $0 < \delta_2 < (2q_- - q_i^{(j)} - q_i^{(j-1)})/3$ be some small perturbations on a quantization level. Then,*

(a)
$$\left|\frac{-\alpha\psi_i(q_+ + \delta_1; \varepsilon)}{\psi_i'(q_+ + \delta_1; \varepsilon)}\right| \leq 2\alpha\delta_1;$$

(b)
$$\left|\frac{-\alpha\psi_i(q_- - \delta_2; \varepsilon)}{\psi_i'(q_- - \delta_2; \varepsilon)}\right| \leq 2\alpha\delta_2.$$

For $j \in \{1, K_i\}$, denote $q_- = q_i^{(1)} - \varepsilon, q_+ = q_i^{(K_i)} + \varepsilon$, and let $\delta_1 > 0, \delta_2 > 0$, we have

$$\left|\frac{-\alpha\psi_i(q_+ + \delta_1; \varepsilon)}{\psi_i'(q_+ + \delta_1; \varepsilon)}\right| = \alpha\delta_1, \left|\frac{-\alpha\psi_i(q_- - \delta_2; \varepsilon)}{\psi_i'(q_- - \delta_2; \varepsilon)}\right| = \alpha\delta_2.$$

An astute reader will notice that δ_1, δ_2 are some small perturbations that attempt to drag an iterate out of the feasible set, while the above inequalities ensure that a skewing force that attempts to correct this perturbation is bounded proportionally by the perturbed iterate's distance to the feasible set. Formally speaking, if we have $\sup_{k\geq k_0} \mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) \leq \delta$ (where δ is small), then $|u_i^{(k)}| \leq 2\alpha\delta$ for all $k \geq k_0$ and every $i \in [d]$.

We leave the proofs of Lemma 3, 4 in Appendix C.

4.1 Convergence Results under Deterministic Full-Batch Gradient Oracles

Now, we consider the case where the gradient obtained by `ASkewSGD` is exact, i.e., $N_b = N$.

We consider the removal of A3, which enables the use of constant step sizes common in machine learning use cases. To derive a non-asymptotic bound, we improved Lemma 2 and obtained a requirement for the learning rate γ. We justify by a counterexample that an upper bound, independent of the Lipschitz smoothness M_ℓ, is necessary for stationarity. With the presence of Lipschitz continuity on the loss function and a small enough learning rate dependent on the time horizon T, we derive a finite-time convergence bound with a convergence rate of $\mathcal{O}(1/\sqrt{T})$.

We summarize our results by the following two theorems.

Theorem 2 (**Finite Time Convergence with Lipschitz Continuity and General Constant Stepsize**). *Let $(\gamma_k)_{k\geq 0}$ be a γ-constant ($\gamma < 1/M_\ell$) sequence. Under A1, A2, we have*

$$\min_{k=0}^{T-1} \|\mathbf{v}^{(k)}\|^2 \leq \frac{2(\ell(\mathbf{w}^{(0)}) - \ell(\mathbf{w}^{(T)}))}{\gamma T} + 2M_c L_\ell d + \gamma M_\ell M_c^2 d + M_c^2 d.$$

with a convergence rate of $\mathcal{O}(1/T)$ and

$$\min_{k=0}^{T-1} \|\mathbf{v}^{(k)}\|^2 \to \mathcal{O}(\gamma + M_c^2 d(1 + L_\ell + M_\ell)),$$

as T tends to infinity.

Theorem 3 (**Finite Time Convergence with Lipschitz Continuity and Small Constant Stepsize**). *For a sufficiently long time horizon T, let $(\gamma_k)_{k\geq 0}$ be a γ-constant sequence, where $\gamma := \sqrt{T} < 1/M_\ell$. Then, under A1, A2, there exists k_1 such that $\mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) \leq \gamma L_\ell \sqrt{d}$ for all $k \geq k_1$ and*

$$\min_{k=k_1}^{k_1+T-1} \|\mathbf{v}^{(k)}\|^2 \leq \frac{2(\ell(\mathbf{w}^{(k_1)}) - \ell(\mathbf{w}^{(k_1+T)}))}{\sqrt{T}} + \frac{4\alpha d L_\ell M_\ell + 8\alpha^2 d L_\ell^2}{\sqrt{T}} + \frac{4\alpha^2 d L_\ell^3}{\sqrt{T^3}}$$

with a convergence rate of $\mathcal{O}(1/\sqrt{T})$, while

$$\min_{k=k_1}^{k_1+T-1} \|\mathbf{v}^{(k)}\|^2 \to 0,$$

as T tends to infinity.

Theorem 3 is a direct consequence of the trade-off between convergence rate and bias. Moreover, Lemma 3 provides us with a convergence certificate as it shows that the best iterate is stationary.

Now, for step sizes that satisfy the Robbins-Monro condition (A3) and only with the Lipschitz smoothness of the loss function (A1), we derive an asymptotic convergence result that bounds the true update direction $\mathbf{v}^{(k)}$, up to an order of $\mathcal{O}(M_c d)$, which is only dependent on the clipping constant M_c and the dimension d. We summarize our results as follows:

Theorem 4 (**Asymptotic Convergence with Lipschitz Smoothness and Robbins-Monro Stepsizes**). *Let $\gamma_k < 1/M_\ell$ for all $k \geq k_0$. Under assumptions A1, A2 and A3, we have*

$$\frac{1}{\sum_{k=k_0}^{T} \gamma_k} \sum_{k=k_0}^{T} \frac{\gamma_k}{2} \|\mathbf{v}^{(k)}\|^2 \to \mathcal{O}(M_c^2 d),$$

as T tends to infinity.

Related proofs are deferred to Appendix D.

4.2 Convergence Results under Stochastic Gradient Oracles

We first impose the following assumption on the notion of "stochasticity":

Assumption 5 (**Unbiasedness and Bounded Variance**). *Let $\{j_1, \ldots, j_{N_b}\}$ where $(N_b \leq N)$ be some random variables. Suppose we define $\widehat{\nabla}\ell(\mathbf{w}) := 1/N_b \sum_{i=1}^{N_b} \nabla \ell_{j_i}(\mathbf{w})$, then the stochastic gradient is unbiased, i.e.*

$$\mathbb{E}_{\xi\sim\pi}[\widehat{\nabla}\ell(\mathbf{x};\xi)] = \nabla\ell(\mathbf{x}), \forall \mathbf{x} \in \mathbb{R}^n,$$

and it has a bounded variance, i.e.

$$\mathbb{E}_{\xi\sim\pi}\left[\left\|\widehat{\nabla}\ell(\mathbf{x};\xi) - \nabla\ell(\mathbf{x})\right\|^2\right] \leq \sigma^2, \forall \mathbf{x} \in \mathbb{R}^n.$$

Now, we are ready to state our result. The idea is close to the classical SGD convergence proof of Ghadimi and Lan's (2013). However, the non-trivial part is to bound the true gradient update when the stochastic gradient is perturbed and flushed by the "skewing force".

Theorem 5 (**Asymptotic Convergence with Lipschitz Continuity and Robbins-Monro Stepsizes**). *Assuming that $0 < \varepsilon \leq \inf_{1\leq i\leq d} \inf_{1\leq j<K_i} |q_i^{(j)} - q_i^{(j+1)}|^2/4$ and A1, A2, A3, A5 holds, where $\{q_i^{(j)}\}$ are the quantization levels, then $\lim_{K\to\infty} \mathrm{d}(\mathbf{w}^{(\tau_K)}, \mathcal{Z}_\varepsilon) = 0$ almost surely, where τ_K is a randomized index at iteration K taking value from k_0 to K, with each value generated with the probability proportional to the stepsizes $(\gamma_k)_{k_0\leq k\leq K}$.*

Proof Sketch. We first consider the descent lemma

$$\mathbb{E}_{\mathbf{w}^{(k)}}[\ell(\mathbf{w}^{(k+1)})] \leq \ell(\mathbf{w}^{(k)}) + \gamma_k \mathbb{E}_{\mathbf{w}^{(k)}}[(\widehat{\mathbf{v}}^{(k)})^\top]\nabla\ell(\mathbf{w}^{(k)}) + \frac{\gamma_k^2 M_\ell}{2}\mathbb{E}_{\mathbf{w}^{(k)}}[\|\widehat{\mathbf{v}}^{(k)}\|^2]$$

and break down the stochastic update coordinate-wise. For coordinates that take the stochastic gradient, they reduce to the normal SGD analysis. Otherwise, if the real update takes a skewing force, by Lemma 2 and 4, we know that it can diminish to an infinitesimally small value over time. Finally, we prove by a limit argument that the remaining coordinates converge to the boundary of the feasible region, which must be stationary for convergence. The randomized index trick is adapted from Ghadimi & Lan (2013). The full proof is deferred to Appendix E.

To give a convergence guarantee with only the Lipschitz smoothness assumption (A1), we employ the Stochastic Approximation (SA) framework. To utilize this framework, we ensure the iterates are bounded by the following assumption, which is common in many supervised learning problems:

Assumption 6 (**Coercivity**). *The function ℓ is coercive, i.e.*

$$\lim_{\|\mathbf{w}\|\to\infty} \ell(\mathbf{w}) \to +\infty.$$

Theorem 6 (**Asymptotic Convergence with Lipschitz Smoothness and Robbins-Monro Stepsizes**). *Assuming that $0 < \varepsilon \leq \inf_{1\leq i\leq d} \inf_{1\leq j<K_i} |q_i^{(j)} - q_i^{(j+1)}|^2/4$, and A1, A3, A5, A6 holds, where $\{q_i^{(j)}\}$ are the quantization levels, then $\lim_{k\to\infty} \mathrm{d}(\mathbf{w}^{(k)}, \mathcal{Z}_\varepsilon) = 0$ almost surely.*

Proof Sketch. We first define the Lyapunov function

$$\mathcal{L}(\mathbf{w}) := \ell(\mathbf{w}) + \frac{\alpha}{2}\sum_{i=1}^{d}[\min(0, \psi_i(\mathbf{w}))]^2,$$

the mean update direction $\mathbf{h}(\mathbf{w}) = \mathbb{E}[\mathbf{s}_{\varepsilon,\alpha}(\widehat{\nabla}\ell(\mathbf{w}), \mathbf{w})|\mathbf{w}]$, and the martingale difference noise $\boldsymbol{\xi}^{(k)} = \widehat{\mathbf{v}}^{(k)} - \mathbf{h}(\mathbf{w}^{(k)})$. Then, we can show that

$$\langle \nabla\mathcal{L}(\mathbf{w}), \mathbf{h}(\mathbf{w})\rangle \leq -\mathtt{b}\|\mathbf{h}(\mathbf{w})\|^2$$

for some $\mathtt{b} > 0$ when $\mathbf{h}(\mathbf{w}) \neq 0$. Lastly, since the iterates are almost surely bounded, $\mathcal{L}$ decreases along the trajectories and the set $\mathcal{Z}_\varepsilon$ is almost surely globally asymptotically stable, we verify the conditions of the Kushner-Clark Theorem and conclude the theorem. The full proof is deferred to Appendix E.

5 NUMERICAL EXPERIMENTS

In this section, we supplement Leconte et al.'s (2023) experimental result by comparing `ASkewSGD` with 3 other approaches: a full precision NN, a NN trained with `BinaryConnect` (Courbariaux et al., 2015), and a NN trained with `ProxQuant` (Bai et al., 2019). We denote [Wx/Ay] as a neural architecture with x-bit precision weights and y-bit activations. For x-bit quantization, we refer to the discrete integer values from -2^{x-1} to $2^{x-1}-1$. The pseudo-code of each method is provided in Appendix F.

Methods. We evaluate the performance of these quantization algorithms on three classical tasks: a non-convex 2D binary classification problem on a shallow neural network, and two image classification benchmarks, one with a simple multi-layer perceptron, the other with a ResNet-18 (He et al., 2015). We consider the quantization of neural network parameters, which reduces the memory usage to store the model, while the quantization of activations is left for future work. For all evaluation that involve quantization on the weight parameters, we apply the function $\text{sign}(\cdot)$ (for [W1/Ay]) or $\text{clamp}(\text{round}(\cdot), -2^{x-1}, 2^{x-1}-1)$ (for [Wx/Ay] with $x \neq 1$) before evaluating it on the test set. We simply do nothing for the full precision [W32/A32] architecture when we refer to "quantized".

Task I - Two Moons Classification We continue to investigate the performance of binary neural networks based on the synthetic "2 moons dataset" presented by Meng et al. (2020) (as illustrated in Figure 3). Our dataset consists of $N = 2000$ training samples (colored in blue and red corresponding to the two classes) and $M = 500$ test samples (colored in black), generated using the `scikit-learn` library's `make_moons` data generator. Two clusters of interlacing half circles will then be generated. We will also add a random Gaussian noise with variance $\sigma = 0.1$ to offset the point from its original position.

We train a binarized 5-layer perception ([W1/A32]) with layer dimensions 2-10-8-4-2 (input-hidden-output, with bias), which serves as the base model (illustrated in Figure 4). All hidden layers use leaky ReLU activation, and weights are initialized with Kaiming normal initialization (Xavier uniform for the output layer). This configuration provides a balanced trade-off between model capacity and computational efficiency for the given task.

For fair comparison, all the methods are trained for $T = 60$ epochs with the learning rate initially set to $lr = 0.05$ (with decay factors of 0.5 applied at epochs 15 and 25) and in batches of 50 samples. We use the

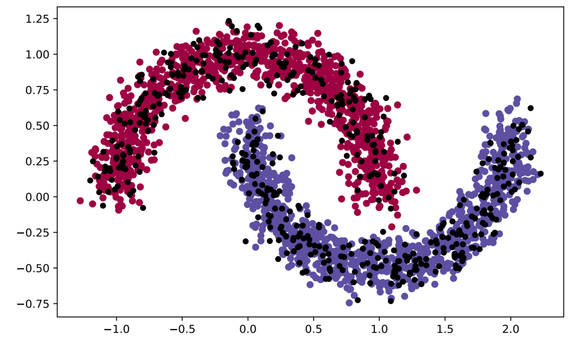

Figure 3: The 2 moons dataset, where the 2000 training samples are colored either blue or red, and the remaining 500 points belonging to the test set are colored black.

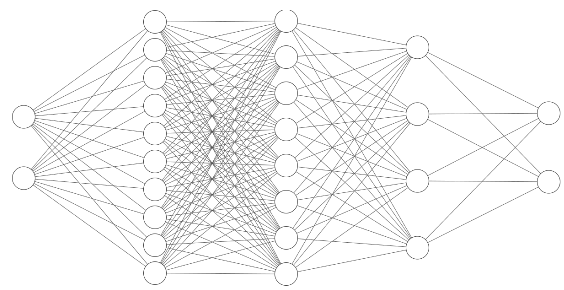

Figure 4: The NN architecture for the 2 moons binary classification task.

`Adam` optimizer (with $\beta_1 = 0.9, \beta_2 = 0.999$) for faster training, and apply the logistic loss function in this experiment. We only apply quantization on the weight matrix (while the bias parameter is kept in full precision for better performance). For `ProxQuant`, we gradually increase the L_2-regularization strength (from epoch 10, and eventually reaches $\lambda = 0.05$ at epoch 60). We start the annealing process for `ASkewSGD` from epoch 21 with a logarithmic schedule $\varepsilon_t = 0.88^{t-20}$ (where t is the epoch) decreasing from epoch to epoch. The hyperparameter α, M_c for `ASkewSGD` is set to 0.2 and 1, respectively.

For this task, we plot the quantized loss per iteration in Figure 5 and report the loss (of the quantized model) of the last iteration in Table 1. Due to regularization, `ProxQuant`'s test loss has temporarily surged for several epochs, followed by stable convergence. We also visualize the decision boundary of each method to provide an intuitive understanding of how each quantization method affects the model's classification behavior (see Figure 6). Results demonstrate that `ProxQuant` and `ASkewSGD` outperform the baseline in maintaining accuracy after quantization, with particular robustness in preserving the model's decision boundaries. The low dimensionality of this task also enabled us to have a better view of the performance of binarized neural networks. Different from Leconte et al.'s (2023) architecture, we increase the number of hidden layers as binarization has limited the development of the expressiveness ability of the original neural network to form any smooth and non-linear decision boundary. Additional results are reported in Appendix F.

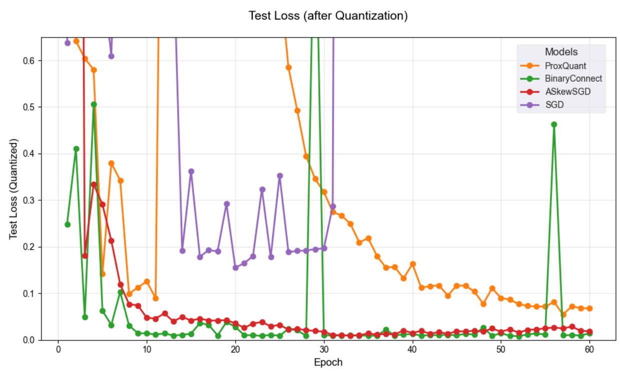

Figure 5: Evaluation of test loss (with weights quantized) on the "2 moons dataset".

Table 1: Performance after 60 epochs (average and standard deviation over 15 random experiments).

Method	**Quantized Test Loss**
Full Precision [W32/A32]	5.44 ± 2.6426
`BinaryConnect` [W1/A32]	$\mathbf{0.03 \pm 0.0731}$
`ProxQuant` [W1/A32]	0.28 ± 0.5932
`ASkewSGD` [W1/A32]	0.05 ± 0.0719

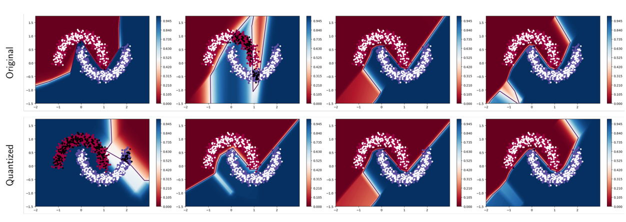

Figure 6: The representative results of the 2 moons classification problem.

Task II - MNIST We investigate the performance of ultra-low-bit quantization on the MNIST handwritten digit classification task (LeCun & Cortes, 2010) for $T = 50$ epochs. The dataset consists of $N = 50000$ training samples (trained in batches of 50) and $M = 10000$ test samples. The preprocessing pipeline is minimal—converting images to tensors and applying dataset-specific normalization ($\mu = 0.1307, \sigma = 0.3081$) - without any geometric transformations or noise injection that are common in modern computer vision pipelines, which enables the focus on the effects of different quantization techniques.

We train a 3-layer MLP model with bias, using a three-layer MLP architecture with 512 hidden units in each layer. The model processes normalized MNIST images (28×28 pixels) through fully connected layers with ReLU activations, culminating in a 10-class output. During the evaluation stage, we quantize both the weights and biases in the model.

The training configuration maintains consistency across methods for fair comparison, using $T = 50$ epochs with a batch size of 200 and initial learning rate of $lr = 0.0002$ for `BinaryConnect` and $lr = 0.06$ for the other three methods. We employ a scheduled learning rate decay ($0.5\times$ reduction at epochs $7, 15, 25$, and 30) that helps stabilize convergence. Multiclass cross-entropy loss function and `SGD` are adopted for all approaches, while each quantization method introduces specialized modifications to the training process.

For 1-bit quantization, we gradually apply L_2 regularization on `ProxQuant` (starting from epoch 20, and reaches $\lambda = 4 \times 10^{-6}$ at epoch 60), while we select $\alpha = 0.004, M_c = 1$ for `ASkewSGD` and anneal with logarithmic decay $\varepsilon = 0.88^{t-20}$ from epoch 20 onward.

For evaluation of multi-bit quantization, we utilize a 2-bit quantization scheme to test the quantization algorithm's robustness. The only difference from the binary quantization case is that we let $\lambda = 4 \times 10^{-5}$ for `ProxQuant` and $\alpha = 0.02, M_c = 1$ for `ASkewSGD`.

We report the final epoch test loss (before quantization is applied) and best quantized test accuracy (over 5 random runs) for all methods in Table 2. The table highlights the trade-offs between full-precision and quantized models, as well as the effectiveness of different quantization techniques at varying bit widths (1-bit and 2-bit weights).

The full-precision model [W32/A32] serves as the baseline, achieving the lowest test loss (0.065 ± 0.001) and the highest quantized accuracy (98.20%). However, `ASkewSGD` [W2/A32] comes remarkably close at 97.55%, achieving competitive performance with low-bit quantization, demonstrating that well-designed gradient updates can mitigate quantization error.

For 1-bit weight quantization [W1/A32], `ASkewSGD` achieves the best quantized accuracy (95.14%) despite exhibiting a high test loss (143.501 ± 6.343), suggesting that its gradient skewing mechanism helps maintain performance despite numerical instability. `ProxQuant` follows closely with 94.94% accuracy and a much lower test loss (0.089 ± 0.001), indicating better optimization stability and consistency between the quantized and non-quantized model. `BinaryConnect` lags be-

Table 2: Performance on MNIST after 50 epochs (average, standard deviation, and best quantized test accuracy) over 5 random experiments.

Method	Test Loss	Best Q. Acc
Full Precision [W32/A32]	**0.065 ± 0.001**	**98.20%**
`BinaryConnect` [W1/A32]	2.289 ± 0.001	93.66%
`ProxQuant` [W1/A32]	0.089 ± 0.001	94.94%
`ASkewSGD` [W1/A32]	143.501 ± 6.343	**95.14%**
`BinaryConnect` [W2/A32]	2.288 ± 0.002	94.80%
`ProxQuant` [W2/A32]	0.144 ± 0.003	93.33%
`ASkewSGD` [W2/A32]	0.068 ± 0.001	**97.55%**

Table 3: Performance of ResNet-18 (best epoch quantized test accuracy, best test loss) on CIFAR-10 after 150 epochs.

Method	Test Loss	Best Q. Acc
Full Precision [W32/A32]	**0.4363**	**88.80%**
`BinaryConnect` [W1/A32]	0.4732	86.24%
`ProxQuant` [W1/A32]	0.4940	83.21%
`ASkewSGD` [W1/A32]	**0.4385**	**87.15%**
`BinaryConnect` [W2/A32]	**0.4271**	87.72%
`ProxQuant` [W2/A32]	0.4760	85.54%
`ASkewSGD` [W2/A32]	0.4508	**87.95%**

hind with 93.66% accuracy, likely due to its deterministic rounding approach, which may introduce larger quantization errors.

When increasing the bit width to 2-bit weights [W2/A32], `ASkewSGD` again leads in quantized accuracy while maintaining a low test loss (0.068 ± 0.001). `BinaryConnect` improves slightly to 94.80%, but `ProxQuant` sees a drop in accuracy (93.33%) compared to its 1-bit version, possibly due to its sensitivity to the regularization parameter λ.

We report the additional results in Appendix F.

Task III - CIFAR-10 We train `ResNet-18` on the CIFAR-10 dataset (Krizhevsky and Hinton, 2009) for $T = 150$ epochs. The CIFAR-10 dataset consists of 60,000 training images and 10,000 test images across 10 classes. During training, we follow the common practices of data augmentation on the training set, including random resizing (to 32×32 pixels with a scale range of 0.64–1.0), horizontal flipping, random erasing (cutout), and normalization using predefined mean and standard deviation values. The test set undergoes only normalization to ensure fair evaluation.

We train a ResNet-18 (He et al., 2015). Quantization is applied to all convolutional weights, while excluding BatchNorm and fully connected layers.

During training, a batch size of 100 is used, with an initial learning rate of 0.06, decayed by a factor of 0.5 at epochs 20 and 40. The `Adam` optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) is employed for all experiments, with cross-entropy loss serving as the training objective.

For `ProxQuant`, we set $\lambda = 1 \times 10^{-4}$ for 1-bit quantization and $\lambda = 0.02$ for 2-bit quantization (starting from epoch 50), while we select $\alpha = 0.2, M_c = 2$ for `ASkewSGD` and anneal with logarithmic decay $\varepsilon = 0.88^{t-90}$ from epoch 90 onward.

We report the best epoch quantized test accuracy and best epoch test loss for all methods in Table 3.

For 1-bit quantization, `ASkewSGD` demonstrates superior performance with 87.15% quantized accuracy - just 1.65% below full precision - and the lowest quantized loss (0.4385) among the other methods. This suggests that its gradient modification strategy effectively preserves network capacity despite extreme weight binarization. `BinaryConnect` follows closely at 86.24%, while `ProxQuant` trails at 83.21%, possibly due to its aggressive regularization scheme.

The 2-bit results show expected improvements, with all methods narrowing the gap to full precision. Notably, `BinaryConnect` achieves the lowest quantized loss (0.4271) in this configuration, though `ASkewSGD` maintains the highest accuracy (87.95%). The consistent superiority of `ASkewSGD` across bit-widths implies its constrained optimization approach adapts well to different quantization levels.

The results collectively demonstrate that: (1) `ASkewSGD`'s balanced approach works best for aggressive quantization, (2) `BinaryConnect` becomes competitive at 2-bit precision, and (3) all methods preserve $> 85\%$ accuracy even with substantial weight compression, validating quantization for efficient deep learning.

6 CONCLUSION

We have observed the dire need for developing novel quantization techniques for deep neural networks. In this paper, we have attempted to investigate the theoretical guarantees provided by a modified variant of `ASkewSGD` to gain insights about how the algorithm has managed to keep the error at a justifiable level. We have also supplemented the experimental results, which aid in our understanding of different quantization algorithms through comparison. Future work should continue on discovering the infinite possibilities of quantization models for filling in the gap in the number of bits required to maintain a good accuracy for neural networks, and develop new theories about the power of the over-parameterization of deep neural network models.

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A THE MANGASARIAN-FROMOVITZ CONSTRAINT QUALIFICATION AND THE CLARKE'S TANGENT CONE

We reproduce the fundamental connection between the Clarke's tangent cone and the quantization constraints for clarity. The fact that we do not have equality constraints in our case has greatly simplified the proof.

Lemma 1. *Let $\mathbf{x} \in \mathbb{R}^d$ be a point in the feasible set C. Then, the Clarke's tangent cone at $\mathbf{x}$, $T(\mathbf{x})$, is equal to the cone $T_M(\mathbf{x})$ as defined in (8).*

Proof. Adapted from Herzog (2023).

$(\Rightarrow)$: $\boldsymbol{\delta}\mathbf{x} \in T(\mathbf{x})$ implies that there exists two sequences $\{\mathbf{x}_j\} \to \mathbf{x}, \{\mathbf{x}_j\} \subset C, t_j \downarrow 0$ for all $j \in \mathbb{N}$ and

$$\frac{\mathbf{x}_j - \mathbf{x}}{t_j} \to \boldsymbol{\delta}\mathbf{x}.$$

Let us fix $i \in I(\mathbf{x})$, then by the Mean-Value Theorem,

$$\begin{aligned} 0 &\leq g_i(\mathbf{x}_j) && \text{(since } \mathbf{x}_j \text{ is feasible)} \\ &= g_i(\mathbf{x}) + \nabla g_i(\mathbf{x} + \xi_j(\mathbf{x}_j - \mathbf{x}))^\top (\mathbf{x}_j - \mathbf{x}) \\ &= 0 + \nabla g_i(\mathbf{x} + \xi_j(\mathbf{x}_j - \mathbf{x}))^\top (\mathbf{x}_j - \mathbf{x}) \end{aligned}$$

for some $\xi_j \in (0, 1)$.

Divide both sides by t_j and take the limit as $j \to \infty$, we have $0 \leq \nabla g_i(\mathbf{x})^\top \boldsymbol{\delta}\mathbf{x}$ wherever $i \in I(\mathbf{x})$.

$(\Leftarrow)$: Let $\boldsymbol{\delta}\mathbf{x} \in T_M(\mathbf{x})$ satisfy $\nabla g_i(\mathbf{x})^\top \boldsymbol{\delta}\mathbf{x} \geq 0, \forall i \in I(\mathbf{x})$, also let $\boldsymbol{\delta}\mathbf{y}$ be given by MFCQ such that $\nabla g_i(\mathbf{w})^\top \boldsymbol{\delta}\mathbf{y} > 0, \forall i \in I(\mathbf{x})$. Put $\boldsymbol{\tau}(t) := \boldsymbol{\delta}\mathbf{x} + t \cdot \boldsymbol{\delta}\mathbf{y}$. Then for all $t > 0$, we have $\nabla g_i(\mathbf{x})^\top \boldsymbol{\tau}(t) > 0, \forall i \in I(\mathbf{x})$, implying that $\boldsymbol{\tau}(t)$ are all feasible MFCQ vectors.

Now, we claim that $\boldsymbol{\tau}(t) \in T(\mathbf{x})$ for all $t \in \mathbb{R}_{++}$. Let $\boldsymbol{\gamma}(u) := \mathbf{x} + u\boldsymbol{\tau}(t)$, $u \in (-\varepsilon, \varepsilon)$, where ε is infinitesimally small given by the continuity of $\mathbf{g}$. Then, $\boldsymbol{\gamma}(u) \in T_M$ for every $u \in [0, \varepsilon)$ and $\boldsymbol{\gamma}(0) = \mathbf{x}, \boldsymbol{\gamma}'(0) = \boldsymbol{\tau}(t)$. For an arbitrary sequence $\{u_j\} \downarrow 0$ and $\mathbf{x}_k = \boldsymbol{\gamma}(u_j) \to \mathbf{x}$ we have

$$\boldsymbol{\tau}(t) = \boldsymbol{\gamma}'(0) = \lim_{j \to \infty} \frac{\boldsymbol{\gamma}(u_j) - \boldsymbol{\gamma}(0)}{u_j - 0} = \lim_{j \to \infty} \frac{\mathbf{x}_j - \mathbf{x}}{t_j} \in T(\mathbf{x}).$$

Since $T(\mathbf{x})$ is closed, $\boldsymbol{\delta}\mathbf{x} = \lim_{t \to 0^+} \boldsymbol{\tau}(t) \in T(\mathbf{x})$. □

B PROOF OF LEMMA 2

Lemma 2. *Assume that A2, A3 holds, we have* $\limsup_{k \to \infty} \mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) = 0$ *almost surely.*

Proof. Reproduced from Leconte et al. (2023).

Denote $M_1 = \max\{M_c, L_\ell\}$. Now, for all $i \in [n]$, we have $\|\widehat{\nabla}\ell(\mathbf{w}^{(k)})\|_2 \leq M_1$ and $|v_i^{(k)}| \leq M_1$, and thus $|w_i^{(k+1)} - w_i^{(k)}| \leq \gamma_k M_1$.

Three crucial steps for the correctness of this lemma will then be introduced. The three steps involved will correspond to Claims 1-3 (confinement to intervals), Claims 4-5 (boundary transition), and Claims 7-8 (convergence to boundary).

Claim 1. For $i \in \{1, 2, \ldots, d\}$, and for $2 \leq j \leq K_i - 1$ if the set $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$ is visited by $w_i^{(k)}$ infinitely often, then there is k_0 such that for all $k > k_0$, $w_i^{(k)} \in [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$.

Fix one such j and denote $[q_-, q_+]$ the set $C_{\varepsilon,i} \cap [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$, where $C_{\varepsilon,i}$ is the projection of C_ε on the i-th coordinate. Define $k_0 = \sup\{k : \gamma_k M_1 \geq \max(q_- - (q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2 - q_+)\}$. Consider $k \geq k_0$, suppose $(q_i^{(j)} + q_i^{(j-1)})/2 \leq w_i^{(k)} < q_-$ (the left side of the interval), then the iterate is pushed to the right and $w_i^{(k)} \leq w_i^{(k+1)}$.

Furthermore, by definition of k_0, it holds that $w_i^{(k+1)} \leq q_- + \gamma_k M_1 \leq (q_i^{(j)} + q_i^{(j+1)})/2$. This implies that $w_i^{(k+1)}$ stays in $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$ in this case.

Otherwise, suppose $q_+ \leq w_i^{(k+1)} < (q_i^{(j)} + q_i^{(j+1)})/2$ (the right side of the interval), then the iterate is pushed to the left and $w_i^{(k)} \geq w_i^{(k+1)} \geq q_+ - \gamma_k M_1 \geq (q_i^{(j)} + q_i^{(j-1)})/2$. Finally, if $w_i^{(k)} \in [q_-, q_+]$, then by the definition of k_0, we obtain $w_i^{(k+1)} \in [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$.

Thus, we have shown that for $k \geq k_0$ if $w_i^{(k)}$ stays within the interval $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$, then for all $k' \geq k$, $w_i^{(k')} \in [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$ by induction.

Similarly, we can prove the two statements below corresponding to the edge cases.

Claim 2. For $i \in \{1, 2, \ldots, d\}$, if the set $(-\infty, (q_i^{(1)} + q_i^{(2)})/2)$ is visited by $w_i^{(k)}$ infinitely often, then there is k_0 such that for all $k > k_0$, $w_i^{(k)} \in (-\infty, (q_i^{(1)} + q_i^{(2)})/2)$.

Claim 3. For $i \in \{1, 2, \ldots, d\}$, if the set $[(q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$ is visited by $w_i^{(k)}$ infinitely often, then there is k_0 such that for all $k > k_0$, $w_i^{(k)} \in [(q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$.

We can then leave our attention to one of the three intervals $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2), (-\infty, (q_i^{(1)} + q_i^{(2)})/2)$, or $[(q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$.

Claim 4. There is $k_1 \geq k_0$, such that if there are two indices $m_+ \geq m_- > k_0$ such that $w_i^{(m_-)} < q_- < q_+ < w_i^{(m_+)}$, then there is m, satisfying $m_- \leq m \leq m_+$, such that $w_i^{(m)} \in [q_-, q_+]$.

Define $k_1 := \sup\{k : \gamma_k M_1 \geq q_+ - q_-\}$. Let m_-, m_+ be given and consider $m := \inf\{k \geq m_- : w_i^{(k)} \geq q_-\}$. It holds that $w_i^{(m-1)} < q_- \leq w_i^{(m)} \leq w_i^{(m-1)} + \gamma_k M_1$.

Since $m \geq k_1$, this implies that $w_i^{(m)} \leq q_- + \gamma_k M_1 \leq q_+$, which proves the claim.

Claim 5. There is $k_1 \geq k_0$, such that if there are two indices $m_- \geq m_+ > k_0$ such that $w_{m_-} < q_- < q_+ < w_{m_+}$, then there is m, satisfying $m_+ \leq m \leq m_-$, such that $w_i^{(m)} \in [q_-, q_+]$.

The proof of Claim 5 is similar to Claim 4. Claims 4, and 5 show that there are only three behaviors of $w_i^{(k)}$. These three conditions will be treated by Claims 6, 7, and 8, respectively.

Claim 6. If $w_i^{(k)}$ visits $[q_-, q_+]$ infinitely often, then $\limsup_{k\to\infty} w_i^{(k)} \leq q_+$ and $\limsup_{k\to\infty} w_i^{(k)} \geq q_-$.

We first show the case of $\limsup_{k\to\infty} w_i^{(k)} \leq q_+$. Suppose that $w_i^{(k)} > q_+$, then $w_i^{(k+1)} \leq w_i^{(k)}$, and otherwise if $w_i^{(k)} \leq q_+$ then $w_i^{(k+1)} \leq q_+ + \gamma_k M_1$. We note that γ_k is non-increasing and thus no iterates $\mathbf{w}^{(k')}$ where $k' \geq k$ at coordinate i can go further than $q_+ + \gamma_k M$ since it immediately receives the reactive pushing force after the constraint violation. Tending k to infinity yields $\limsup_{k\to\infty} w_i^{(k)} \leq c_+$. It is easy to see that the other case also holds.

Claim 7. If for all k large enough, $w_i^{(k)} > q_+$, then $w_i^{(k)} \to q_+$.

For all k large enough, the sequence $(w_i^{(k)})_{k\geq k_0}$ receives a pushing force from the right and thus is decreasing. From the given condition that $w_i^{(k)} > q_+$, $w_i^{(k)}$ is decreasing and bounded below, thus the sequence has a limit. Assume the contrary that $\lim_{k\to\infty} w_i^{(k)} \neq q_+$. It holds that $w_i^{(k+m+1)} \leq w_i^{(k)} - M_+ \sum_{j=0}^{m} \gamma_{k+j}$, where $M_+ = \inf\{\min(M_{\varepsilon,c}, \alpha|\psi_i(w_i; \varepsilon)|/\psi_i'(w_i; \varepsilon)) : w_i \in [q_+, (q_i^{(j)} + q_i^{(j+1)})/2)\} > 0$. Since $\sum_{j=0}^{\infty} \gamma_j = +\infty$, we have reached a contradiction. $w_i^{(k)} \to q_+$.

Claim 8. If for all k large enough, $w_i^{(k)} < q_-$, then $w_i^{(k)} \to q_-$.

It holds that $w_i^{(k+m+1)} \geq w_i^{(k)} + M_- \sum_{j=0}^{m} \gamma_{k+j}$, where $M_- = \inf\{\min(M_{\varepsilon,c}, \alpha|\psi_i(w_i; \varepsilon)|/\psi_i'(w_i; \varepsilon)) : w_i \in ((q_i^{(j)} + q_i^{(j-1)})/2, q_-]\} > 0$. Similar to Claim 7, it is impossible that $\lim_{k\to\infty} w_i^{(k)} \neq q_-$. $\square$

C PROOF OF LEMMA 3 AND 4

The following lemma shows that $\mathbf{v} := \mathbf{s}_{\varepsilon,\alpha}(\nabla \ell(\mathbf{w}), \mathbf{w})$ can be a surrogate optimality indicator for $(\mathcal{P}_\varepsilon)$.

Lemma 3. *$\mathbf{w} \in \mathcal{Z}_\varepsilon$ if and only if $\mathbf{s}_{\varepsilon,\alpha}(\nabla \ell(\mathbf{w}), \mathbf{w}) = \mathbf{0}$.*

Proof. Denote $\mathbf{h}(\mathbf{w}) = \mathbb{E}[\mathbf{s}_{\varepsilon,\alpha}(\widehat{\nabla} \ell(\mathbf{w}), \mathbf{w})] = \mathbf{s}_{\varepsilon,\alpha}(\nabla \ell(\mathbf{w}), \mathbf{w})$. We first prove the "only-if" direction. When $\mathbf{w} \in \mathcal{Z}_\varepsilon$ (which is a subset of C_ε), $\psi_i(w_i; \varepsilon) \geq 0$ for all $i \in [d]$. If $\psi_i(w_i; \varepsilon) > 0$, then $[\nabla \ell(\mathbf{w})]_i = 0$. Otherwise, $\alpha \psi_i(w_i; \varepsilon)/\psi_i'(w_i; \varepsilon) = 0$ and hence we just need to consider the case where $-[\nabla \ell(\mathbf{w})]_i \cdot \psi_i'(w_i; \varepsilon) \geq 0$, this is impossible as $\operatorname{sign}([\nabla \ell(\mathbf{w})]_i) = \operatorname{sign}(\psi_i'(w_i; \varepsilon))$.

We then investigate the "if" direction. If $\psi_i(w_i; \varepsilon) > 0$, we can make use of the fact that $[\nabla \ell(\mathbf{w})]_i = [\mathbf{h}(\mathbf{w})]_i = 0$. If $\psi_i(w_i; \varepsilon) < 0$ and $-[\nabla \ell(\mathbf{w})]_i \cdot \psi_i'(w_i; \varepsilon) \leq \alpha \psi_i(w_i; \varepsilon)$, then $[\mathbf{h}(\mathbf{w})]_i \neq \mathbf{0}$, which is a contradiction. This implies that $0 = -[\nabla \ell(\mathbf{w})]_i \cdot \psi_i'(w_i; \varepsilon) \geq \alpha \psi_i(w_i; \varepsilon) > 0$, which is also impossible. If $\psi_i(w_i; \varepsilon) = 0$ and $-[\nabla \ell(\mathbf{w})]_i \cdot \psi_i'(w_i; \varepsilon) \leq \alpha \psi_i(w_i; \varepsilon) = 0$, then indeed we have $\operatorname{sign}([\nabla \ell(\mathbf{w})]_i) = \operatorname{sign}(\psi_i'(w_i; \varepsilon))$. Now, we are left with the conditions that takes $[\mathbf{h}(\mathbf{w})]_i = -[\nabla \ell(\mathbf{w})]_i = 0$ and $\psi_i(w_i; \varepsilon) \geq 0$, which is always satisfactory. $\square$

Lemma 4. *Fix an arbitrary $0 < \varepsilon \leq \inf_{1 \leq i \leq d} \inf_{1 \leq j < K_i} |q_i^{(j)} - q_i^{(j+1)}|^2/4$, where $\{q_i^{(j)}\}$ are the quantization levels. For $j \in \{2, \ldots, K_i - 1\}$, denote $[q_-, q_+]$ as the set $C_{\varepsilon,i} \cap [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$, where $C_{\varepsilon,i}$ is the projection of C_ε on the i-th coordinate. Let $0 < \delta_1 < (q_i^{(j)} + q_i^{(j+1)} - 2q_+)/3$ and $0 < \delta_2 < (2q_- - q_i^{(j)} - q_i^{(j-1)})/3$ be some small perturbations on a quantization level. Then,*

(a)

$$\left| \frac{-\alpha \psi_i(q_+ + \delta_1; \varepsilon)}{\psi_i'(q_+ + \delta_1; \varepsilon)} \right| \leq 2\alpha\delta_1;$$

(b)

$$\left| \frac{-\alpha \psi_i(q_- - \delta_2; \varepsilon)}{\psi_i'(q_- - \delta_2; \varepsilon)} \right| \leq 2\alpha\delta_2.$$

For $j \in \{1, K_i\}$, denote $q_- = q_i^{(1)} - \varepsilon, q_+ = q_i^{(K_i)} + \varepsilon$, and let $\delta_1 > 0, \delta_2 > 0$, we have

$$\left| \frac{-\alpha \psi_i(q_+ + \delta_1; \varepsilon)}{\psi_i'(q_+ + \delta_1; \varepsilon)} \right| = \alpha\delta_1, \left| \frac{-\alpha \psi_i(q_- - \delta_2; \varepsilon)}{\psi_i'(q_- - \delta_2; \varepsilon)} \right| = \alpha\delta_2.$$

Proof. By symmetry, we only need to prove the statement as in (a). Note that

$$\psi_i(q_+; \varepsilon) = \varepsilon - (q_+ - q_i^{(j-1)})(q_i^{(j)} - q_+) = 0 \tag{14}$$

and

$$\psi_i'(w; \varepsilon) = 2w - q_i^{(j+1)} - q_i^{(j)}. \tag{15}$$

Using (14), we expand the numerator of the left-hand side of (a) and obtain

$$\begin{aligned} \psi_i(q_+ + \delta_1; \varepsilon) &= \varepsilon + (q_+ + \delta_1 - q_i^{(j)})(q_+ + \delta_1 - q_i^{(j+1)}) & (16)\\ &= \varepsilon + (q_+ - q_i^{(j)})(q_+ - q_i^{(j+1)}) + \delta_1(2q_+ - q_i^{(j+1)} - q_i^{(j)}) + \delta_1^2 & (17)\\ &= \delta_1(2q_+ - q_i^{(j+1)} - q_i^{(j)}) + \delta_1^2 < 0. & (18) \end{aligned}$$

Since $\delta_1 > 0$, we have $\psi_i'(q_+ + \delta_1; \varepsilon) = 2q_+ + \delta_1 - q_i^{(j+1)} - q_i^{(j)} < 0$ by (15). Then,

$$\begin{aligned} \left| \frac{-\alpha \psi_i(q_+ + \delta_1; \varepsilon)}{\psi_i'(q_+ + \delta_1; \varepsilon)} \right| &= \frac{\alpha\delta_1(2q_+ - q_i^{(j)} - q_i^{(j+1)} + \delta_1)}{(2q_+ - q_i^{(j)} - q_i^{(j+1)}) + 2\delta_1} & (19)\\ &\leq \frac{\alpha\delta_1(2q_+ - q_i^{(j)} - q_i^{(j+1)})}{(2q_+ - q_i^{(j)} - q_i^{(j+1)}) + 2\delta_1} & (20)\\ &= \alpha\delta_1 - \frac{\alpha\delta_1^2}{(2q_+ - q_i^{(j)} - q_i^{(j+1)}) + 2\delta_1}. & (21) \end{aligned}$$

Since $\delta_1 \le (q_i^{(j)} + q_i^{(j+1)} - 2q_+)/3$, we have

$$\left| \frac{-\alpha\psi_i(q_+ + \delta_1; \varepsilon)}{\psi_i'(q_+ + \delta_1; \varepsilon)} \right| \le 2\alpha\delta_1. \tag{22}$$

The edge cases where $j \in \{1, K_i\}$ should be apparent by simple calculations. □

D PROOF OF THEOREM 2 AND 3

In the sequel, we present our convergence result of `ASkewSGD` with deterministic gradient oracles. We first prove a convergence bound when general constant step size is employed. We categorize the different types of updates coordinate-wise.

Definition 4 (**Categorization of the Update Direction**). *For the update on the i-th coordinate, we define*

$$i \in S_\varepsilon(\mathbf{w}) \Longleftrightarrow \psi_i(w_i;\varepsilon) \le 0, \tag{23}$$
$$i \notin S_\varepsilon(\mathbf{w}) \Longleftrightarrow \psi_i(w_i;\varepsilon) > 0 \text{ or } -[\nabla\ell(\mathbf{w})]_i \cdot \psi_i'(w_i;\varepsilon) \ge -\alpha\psi_i(w_i;\varepsilon) \ge 0. \tag{24}$$

Theorem 2 (Finite Time Convergence with Lipschitz Continuity and General Constant Stepsize). *Let $(\gamma_k)_{k\ge 0}$ be a γ-constant ($\gamma < 1/M_\ell$) sequence. Under A1, A2, we have*

$$\min_{k=0}^{T-1} \|\mathbf{v}^{(k)}\|^2 \le \frac{2(\ell(\mathbf{w}^{(0)}) - \ell(\mathbf{w}^{(T)}))}{\gamma T} + 2M_c L_\ell d + \gamma M_\ell M_c^2 d + M_c^2 d$$

with a convergence rate of $\mathcal{O}(1/T)$ and

$$\min_{k=0}^{T-1} \|\mathbf{v}^{(k)}\|^2 \to \mathcal{O}(\gamma + M_c^2 d(1 + L_\ell + M_\ell)),$$

as T tends to infinity.

Proof. First, apply the descent lemma on the M_ℓ-smooth function ℓ. Then, we discuss the two cases of the update coordinate-wise as in Definition 4. We obtain the following inequality:

$$\ell(\mathbf{w}^{(k+1)}) \le \ell(\mathbf{w}^{(k)}) + \gamma(\mathbf{v}^{(k)})^\top \nabla\ell(\mathbf{w}^{(k)}) + \frac{\gamma^2 M_\ell}{2}\|\mathbf{v}^{(k)}\|^2 \tag{25}$$

$$= \ell(\mathbf{w}^{(k)}) + \gamma \sum_{i\in S_\epsilon(\mathbf{w}^{(k)})} u_i^{(k)}[\nabla\ell(\mathbf{w}^{(k)})]_i - \gamma \sum_{i\notin S_\epsilon(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2$$
$$+ \frac{\gamma^2 M_\ell}{2} \sum_{i\in S_\epsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2 + \frac{\gamma^2 M_\ell}{2} \sum_{i\notin S_\epsilon(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^k)]_i^2. \tag{26}$$

Since $\gamma < 1/M_\ell$, we have $1 - \gamma M_\ell/2 > 1/2$, then

$$\ell(\mathbf{w}^{(k+1)}) \le \ell(\mathbf{w}^{(k)}) - \frac{\gamma}{2} \sum_{i\notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2 + \gamma \sum_{i\in S_\varepsilon(\mathbf{w}^{(k)})} u_i^{(k)}[\nabla\ell(\mathbf{w}^{(k)})]_i + \frac{\gamma^2 M_\ell}{2} \sum_{i\in S_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2. \tag{27}$$

Since the gradient of ℓ is bounded by L_ℓ and $|u_i^{(k)}|$ is bounded by M_c, we have

$$\ell(\mathbf{w}^{(k+1)}) \le \ell(\mathbf{w}^{(k)}) - \frac{\gamma}{2} \sum_{i\notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2 + \gamma \sum_{i\in S_\epsilon(\mathbf{w}^{(k)})} M_c L_\ell + \frac{\gamma^2 M_\ell}{2} \sum_{i\in S_\varepsilon(\mathbf{w}^{(k)})} M_c^2 \tag{28}$$

$$\le \ell(\mathbf{w}^{(k)}) - \frac{\gamma}{2} \sum_{i\notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2 + \gamma M_c L_\ell d + \frac{\gamma^2 M_\ell M_c^2 d}{2}. \tag{29}$$

Rearrange the inequality and summing over T epochs from 0 to $T-1$, we obtain

$$\frac{\gamma}{2}\sum_{k=0}^{T-1}\sum_{i\notin S_\varepsilon(\mathbf{w}^{(k)})}[\nabla\ell(\mathbf{w}^{(k)})]_i^2 \le \ell(\mathbf{w}^{(0)})-\ell(\mathbf{w}^{(T)})+T\gamma M_c L_\ell d + T\frac{\gamma^2 M_\ell M_c^2 d}{2}. \tag{30}$$

Completing the entire update direction with the skewing force, we can bound $\|\mathbf{v}^{(k)}\|^2$ as we have

$$\begin{aligned}\frac{\gamma}{2}\sum_{k=0}^{T-1}\left(\sum_{i\notin S_\varepsilon(\mathbf{w}^{(k)})}[\nabla\ell(\mathbf{w}^{(k)})]_i^2+\sum_{i\in S_\varepsilon(\mathbf{w}^{(k)})}(u_i^{(k)})^2\right) &\le \ell(\mathbf{w}^{(0)})-\ell(\mathbf{w}^{(T)})+T\gamma M_c L_\ell d \\ &\quad + T\frac{\gamma^2 M_\ell M_c^2 d}{2}+\frac{\gamma}{2}TM_c^2 d,\end{aligned} \tag{31}$$

with

$$\|\mathbf{v}^{(k)}\|^2=\sum_{i\notin S_\varepsilon(\mathbf{w}^{(k)})}[\nabla\ell(\mathbf{w}^{(k)})]_i^2+\sum_{i\in S_\varepsilon(\mathbf{w}^{(k)})}(u_i^{(k)})^2, \tag{32}$$

and hence

$$\begin{aligned}\min_{k=0}^{T-1}\left(\sum_{i\notin S_\varepsilon(\mathbf{w}^{(k)})}[\nabla\ell(\mathbf{w}^{(k)})]_i^2+\sum_{i\in S_\varepsilon(\mathbf{w}^{(k)})}(u_i^{(k)})^2\right) &\le \frac{2(\ell(\mathbf{w}^{(0)})-\ell(\mathbf{w}^{(T)}))}{\gamma T}+2M_c L_\ell d \\ &\quad + \gamma M_\ell M_c^2 d + M_c^2 d.\end{aligned} \tag{33}$$

Finally,

$$\min_{k=0}^{T-1}\|\mathbf{v}^{(k)}\|^2 \le \frac{2(\ell(\mathbf{w}^{(0)})-\ell(\mathbf{w}^{(T)}))}{\gamma T}+2M_c L_\ell d+\gamma M_\ell M_c^2 d+M_c^2 d. \tag{34}$$

As T tends to infinity, we have

$$\min_{k=0}^{T-1}\|\mathbf{v}^{(k)}\|^2 \to \mathcal{O}(\gamma+M_c^2 d(1+L_\ell+M_\ell)), \tag{35}$$

with convergence rate of $\mathcal{O}(1/T)$. □

We have an extra M_c term as we cannot be sure whether $w_i^{(k)}$ will converge to $C_{\varepsilon,i}$. Large gradient could disrupt the distance to feasible set for general γ. Now, we show that it is impossible to remove the M_c term without upper-bounding γ by an illustrative example shown below:

Example 1. *We consider the case where $\mathcal{Q}=\{\pm 1\}$ and ℓ is decreasing away from $(-1,1)$. Let $[q_-,q_+]=\{w:\psi(w;\varepsilon)\ge 0\}$. Let $w^{(0)}<q_-$. There exists $\gamma=2w^{(0)}/(\alpha(\varepsilon+w^{(0)}+1)))$ such that $w^{(k)}$ enters a cycle.*

To see this, first note that $\psi(w^{(0)};\varepsilon)=\varepsilon-(-1-w^{(0)})<0$. We compute that the skewing correction is

$$u^{(0)}=\frac{-\alpha\psi(w^{(0)})}{\psi'(w^{(0)})}=-\alpha(\varepsilon+w^{(0)}+1).$$

Then, $w^{(1)}=w^{(0)}+\gamma u^{(0)}=w^{(0)}-\gamma\cdot\alpha(\varepsilon+w^{(0)}+1))=w^{(0)}-2w^{(0)}=-w^{(0)}$.

By symmetry, we have $w^{(2)}=-w^{(1)}=w^{(0)}$. Henceforth, by induction we can verify that $w^{(k)}$ ends up in a 2-cycle as a large γ that prevents $w^{(k)}$ from entering the feasible region.

In the sequel, we introduce three lemmata that provides non-asymptotic guarantees for ASkewSGD under Assumption A2 and a small enough step size γ (which will be regulated by the lemmata).

Lemma 5 (**Confinement to the Quantization Interval, General Case**)**.** *Assume that A2 holds. Fix an $i\in[d]$ and $2\le j\le K_i-1$. Let $[q_-,q_+]=C_{\varepsilon,i}\cap[(q_i^{(j-1)}+q_i^{(j)})/2,(q_i^{(j)}+q_i^{(j+1)})/2)$, where $C_{\varepsilon,i}$ is the projection of C_ε on the i-th coordinate, and $0<\gamma\cdot\max\{L_\ell,M_c\}<\min\{q_+-q_-,q_--(q_i^{(j-1)}+q_i^{(j)})/2,(q_i^{(j-1)}+q_i^{(j)})/2-q_+\}$. If there is a moment k_0 such that $w_i^{(k_0)}\in((q_i^{(j)}+q_i^{(j-1)})/2,(q_i^{(j)}+q_i^{(j+1)})/2)$, then $w_i^{(k)}$ is confined to the interval $((q_i^{(j)}+q_i^{(j-1)})/2,(q_i^{(j)}+q_i^{(j+1)})/2)$ for every $k\ge k_0$.*

Proof. Notice for any $k > 0$, $\|\nabla \ell(\mathbf{w}^{(k)})\| \le L_\ell$ and $|v_i^{(k)}| \le \max\{L_\ell, M_c\}$. If $(q_i^{(j)} + q_i^{(j-1)})/2 \le w_i^{(k_0)} < q_-$, then $w_i^{(k_0)}$ is pushed to the right. We have $w_i^{(k_0)} \le w_i^{(k_0+1)} := w_i^{(k_0)} + \gamma \max\{L_\ell, M_c\} < q_+$. If $q_+ < w_i^{(k)} < (q_i^{(j)} + q_i^{(j+1)})/2$, then $w_i^{(k_0)}$ is pushed to the left, and $w_i^{(k_0)} \ge w_i^{(k_0+1)} := w_i^{(k_0)} - \gamma \max\{M_\ell, M_c\} \le q_-$. Finally, if $w_i^{(k_0)} \in [q_-, q_+]$, then $(q_i^{(j-1)} + q_i^{(j)})/2 < w_i^{(k_0)} - \gamma L_\ell \le w_i^{(k_0+1)} \le w_i^{(k_0)} + \gamma L_\ell < (q_i^{(j)} + q_i^{(j+1)})/2$. The same argument is inductively true for all $k \ge k_0$, thus concluding the proof. □

Lemma 6 (**Confinement to the Quantization Interval, Edge Case**). *Assume that A2 holds. Fix an $i \in [d]$. Let $[q_-, q_+] = C_{\varepsilon,i} \cap (-\infty, (q_i^{(1)} + q_i^{(2)})/2)$, where $C_{\varepsilon,i}$ is the projection of C_ε on the i-th coordinate, and $0 < \gamma \cdot \max\{L_\ell, M_c\} < \min\{q_+ - q_-, (q_i^{(1)} + q_i^{(2)})/2 - q_+\}$. If there is a moment k_0 such that $w_i^{(k_0)} \in (-\infty, (q_i^{(1)} + q_i^{(2)})/2)$, then $w_i^{(k)}$ is confined to the interval $(-\infty, (q_i^{(1)} + q_i^{(2)})/2)$ for every $k \ge k_0$. The same conclusion holds for $[q_-, q_+] = C_{\varepsilon,i} \cap ((q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$ and $0 < \gamma \cdot \max\{L_\ell, M_c\} < \min\{q_+ - q_-, (q_i^{(K_i-1)} + q_i^{(K_i)})/2 - q_-\}$ with $w_i^{(k)}$ being confined to the interval $((q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$ for every $k \ge k_0$.*

Proof. Similar to Lemma 3. If $w_i^{(k_0)} < q_-$, then $w_i^{(k_0)}$ is pushed to the right. We have $w_i^{(k_0)} \le w_i^{(k_0+1)} := w_i^{(k_0)} + \gamma \max\{L_\ell, M_c\} < q_+$. If $w_i^{(k_0)} \in [q_-, q_+]$, then $w_i^{(k_0+1)} \le w_i^{(k_0)} + \gamma L_\ell < (q_i^{(1)} + q_i^{(2)})/2$. □

Lemma 7 (**Convergence to Neighborhood of The Feasible Region**). *Assume that A2 holds. Let γ be such that each coordinate i of $\mathbf{w}^{(k_0)}$ is confined to an interval I_i as in Lemma 5 and 6. Let $\tau = \min_i \inf_{\mathrm{d}(w_i, C_{\varepsilon,i}) \ge \gamma L_\ell} \{\mathrm{clip}(|\alpha \psi_i(w_i; \varepsilon)|/|\psi_i'(w_i; \varepsilon)|, M_c)\}$. Then, $\mathrm{d}(w_i^{(k)}, C_{\varepsilon,i}) \le \gamma L_\ell$ for all $k \ge \max\{k_0, k_0 + \lceil (\mathrm{d}(w_i^{(k_0)}, C_{\varepsilon,i}) - \gamma L_\ell)/(\eta\gamma) \rceil\}$, where $C_{\varepsilon,i}$ is the projection of C_ε on the i-th coordinate.*

Proof. Consider the case where $w_i^{(k)} \in C_{\varepsilon,i}$. After an update, $\mathrm{d}(w_i^{(k+1)}, C_{\varepsilon,i}) \le \gamma L_\ell$. Otherwise, the skewing force activates at $k+1$. For each infeasible coordinate i, the skewing correction is at least $\tau\gamma$ by definition (since the skewing force diminishes as $w_i^{(k)}$ gets closer to the feasible region, and the gradient pushes at least as strong as the skewing force). Hence, $\mathrm{d}(w_i^{(k+1)}, C_{\varepsilon,i}) \le \mathrm{d}(w_i^{(k)}, C_{\varepsilon,i}) - \tau\gamma$. Therefore, in at most an extra $k_1 = \max\{k_0, k_0 + \lceil (\mathrm{d}(w_i^{(k_0)}, C_{\varepsilon,i}) - \gamma L_\ell)/(\tau\gamma) \rceil\}$ steps, we can guarantee that the distance to the feasible set is $\mathrm{d}(\mathbf{w}^{(k_1)}, C_\varepsilon) \le \sqrt{\sum_i (\gamma L_\ell)^2} \le \gamma L_\ell \sqrt{d}$, which is at an order of $\mathcal{O}(\gamma)$. □

To simplify matters, the following corollary provides us with a principled way to select a "small" enough step size.

Corollary 1 (Bounding Intervals and Selection of Stepsize). *Assume that A2 holds. Then, for any $\mathbf{w}^{(k_0)}$, there exists γ small enough and k_1 such that $\mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) \le \gamma L_\ell \sqrt{d} =: \delta$ and $|u_i^{(k)}| \le 2\alpha\delta$ for all $k \ge k_1 \ge k_0$ and $i \in [d]$.*

Proof. Let $M = \{L_\ell, M_c\}$ and $\tau = \min_i \inf_{\mathrm{d}(w_i, C_{\varepsilon,i}) \ge \gamma L_\ell} \{(|\alpha \psi_i(w_i; \varepsilon)|/|\psi_i'(w_i; \varepsilon)|, M_c)\}$.

Consider the i-th coordinate of $\mathbf{w}^{(k)}$. There are three cases to consider:

(a) If $w_i^{(k)} < (q_i^{(1)} + q_i^{(2)})/2$, then let $[a_i, b_i] = C_{\varepsilon,i} \cap [-\infty, (q_i^{(1)} + q_i^{(2)})/2))$, we set $\eta_i := \sup\{\gamma : \gamma M < \min\{q_+ - q_-, (q_i^{(1)} + q_i^{(2)})/2 - q_+\}\}$;

(b) If $w_i^{(k)} \ge (q_i^{(K_i-1)} + q_i^{(K_i)})/2)$, then let $[a_i, b_i] = C_{\varepsilon,i} \cap [(q_i^{(K_i-1)} + q_i^{(K_i)})/2), +\infty)$, we set $\eta_i := \sup\{\gamma : \gamma M < \min\{q_+ - q_-, q_- - (q_i^{(K_i-1)} + q_i^{(K_i)})/2\}\}$;

(c) If $(q_i^{(j-1)} + q_i^{(j)})/2 \le w_i^{(k)} < (q_i^{(j)} + q_i^{(j+1)})/2$, then let $[a_i, b_i] = C_{\varepsilon,i} \cap [(q_i^{(j-1)} + q_i^{(j)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$, we set $\eta_i := \min\{\sup\{\gamma : \gamma M < \min\{q_+ - q_-, q_- - (q_i^{(j-1)} + q_i^{(j)})/2, (q_i^{(j-1)} + q_i^{(j)})/2 - q_+\}\}, (q_i^{(j)} + q_i^{(j+1)} - 2q_+)/3, (2q_- - q_i^{(j)} - q_i^{(j-1)})/3\}$.

Then, the proof is immediately concluded by considering the selection of any γ that satisfies $0 < \gamma < \min_i \eta_i$ and $k_1 := \max\{k_0, k_0 + \lceil (\mathrm{d}(\mathbf{w}^{(k_0)}, C_\varepsilon) - \gamma L_\ell)/(\tau\gamma) \rceil\}$, which in fact satisfy Lemma 5, 6 and 7. □

Theorem 7 (**Finite Time Convergence with Lipschitz Continuity and Small Constant Stepsize**). *Let $(\gamma_k)_{k \ge 0}$ be a γ-constant sequence, where $\gamma < 1/L_\ell$ and k_1 are chosen as in Corollary 1. Then, under assumption A1, A2, we have*

$$\min_{k=k_1}^{k_1+T} \|\mathbf{v}^{(k)}\|^2 \le \frac{2(\ell(\mathbf{w}^{(k_1)}) - \ell(\mathbf{w}^{(k_1+T+1)}))}{\gamma(T+1)} + 4\alpha d\gamma L_\ell M_\ell + 4\alpha^2 d\gamma^3 L_\ell^3 + 8\alpha^2 d\gamma L_\ell^2.$$

with a convergence rate of $\mathcal{O}(1/T)$ *and*

$$\min_{k=k_1}^{k_1+T} \|\mathbf{v}^{(k)}\|^2 \to \mathcal{O}(\gamma),$$

as T *tends to infinity.*

Proof. We start from the descent inequality (27) obtained previously

$$\ell(\mathbf{w}^{(k+1)}) \leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma \sum_{i \in S_\varepsilon(\mathbf{w}^{(k)})} u_i^{(k)} [\nabla \ell(\mathbf{w}^{(k)})]_i + \frac{\gamma^2 M_\ell}{2} \sum_{i \in S_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2. \tag{36}$$

Since the gradient of ℓ is bounded by M_ℓ and $|u_i^{(k)}| \leq 2\alpha\gamma L_\ell$ as a consequence of Corollary 1, we have

$$\begin{aligned}
\ell(\mathbf{w}^{(k+1)}) &\leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma \sum_{i \in S_\epsilon(\mathbf{w}^{(k)})} (2\alpha\gamma L_\ell) M_\ell + \frac{\gamma^2 L_\ell}{2} \sum_{i \in S_\varepsilon(\mathbf{w}^{(k)})} (2\alpha\gamma L_\ell)^2 &(37)\\
&\leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma(2\alpha\gamma L_\ell) M_\ell d + \frac{\gamma^2 L_\ell}{2} (2\alpha\gamma L_\ell)^2 d &(38)\\
&\leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + 2\alpha d \gamma^2 M_\ell L_\ell + 2\alpha^2 d \gamma^4 L_\ell^3. &(39)
\end{aligned}$$

Rearrange the inequality and summing over T epochs from k_1 to $k_1 + T - 1$, we obtain

$$\frac{\gamma}{2} \sum_{k=k_1}^{k_1+T-1} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 \leq \ell(\mathbf{w}^{(k_1)}) - \ell(\mathbf{w}^{(k_1+T)}) + 2T\alpha d\gamma^2 L_\ell M_\ell + 2T\alpha^2 d\gamma^4 L_\ell^3. \tag{40}$$

Completing the entire update direction with the skewing force, we can bound $\|\mathbf{v}^{(k)}\|$ as we have

$$\begin{aligned}
\frac{\gamma}{2} \sum_{k=k_1}^{k_1+T-1} \left(\sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \sum_{i \in S_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2 \right) &\leq \ell(\mathbf{w}^{(k_1)}) - \ell(\mathbf{w}^{(k_1+T)}) + 2T\alpha d\gamma^2 L_\ell M_\ell \\
&\quad + 2T\alpha^2 d\gamma^4 L_\ell^3 + 4T\alpha^2 d\gamma^2 L_\ell^2 &(41)
\end{aligned}$$

Finally, as in Equation (32), we have

$$\min_{k=k_1}^{k_1+T-1} \|\mathbf{v}^{(k)}\|^2 \leq \frac{2(\ell(\mathbf{w}^{(k_1)}) - \ell(\mathbf{w}^{(k_1+T)}))}{\gamma T} + 4\alpha d\gamma L_\ell M_\ell + 4\alpha^2 d\gamma^3 L_\ell^3 + 8\alpha^2 d\gamma L_\ell^2. \tag{42}$$

As T tends to infinity, we have

$$\min_{k=k_1}^{k_1+T-1} \|\mathbf{v}^{(k)}\|^2 \to \mathcal{O}(\gamma), \tag{43}$$

with convergence rate of $\mathcal{O}(1/T)$. ☐

Theorem 3 (Time Horizon Dependent Convergence with Lipschitz Continuity and Small Constant Stepsize). *For a sufficiently long time horizon* T, *let* $(\gamma_k)_{k\geq 0}$ *be a* γ*-constant sequence, where* $\gamma := \sqrt{T} < 1/M_\ell$. *Then, under A1, A2, there exists* k_1 *such that* $\mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) \leq \gamma L_\ell \sqrt{d}$ *for all* $k \geq k_1$ *and*

$$\min_{k=k_1}^{k_1+T-1} \|\mathbf{v}^{(k)}\|^2 \leq \frac{2(\ell(\mathbf{w}^{(k_1)}) - \ell(\mathbf{w}^{(k_1+T)}))}{\sqrt{T}} + \frac{4\alpha d L_\ell M_\ell + 8\alpha^2 d L_\ell^2}{\sqrt{T}} + \frac{4\alpha^2 d L_\ell^3}{\sqrt{T^3}}$$

with a convergence rate of $\mathcal{O}(1/\sqrt{T})$, *while*

$$\min_{k=k_1}^{k_1+T-1} \|\mathbf{v}^{(k)}\|^2 \to 0,$$

as T *tends to infinity.*

Proof. Immediate from Corollary 1 and Theorem 7 with the substitution $\gamma = \sqrt{T}$. ☐

Theorem 4 (Asymptotic Convergence with Lipschitz Smoothness and Robbins-Monro Stepsizes). *Let $\gamma_k < 1/M_\ell$ for all $k \geq k_0$. Under assumptions A1, A2 and A3, we have*

$$\frac{1}{\sum_{k=k_0}^{T} \gamma_k} \sum_{k=k_0}^{T} \frac{\gamma_k}{2} \|\mathbf{v}^{(k)}\|^2 \to \mathcal{O}(M_c^2 d),$$

as T tends to infinity.

Proof. Other than Definition 4, we also define

$$i \in S_\varepsilon^+(\mathbf{w}) \iff \psi_i(w_i;\varepsilon) \leq 0 \text{ and } -\alpha\psi_i(w_i;\varepsilon) > -[\nabla \ell(\mathbf{w})]_i \cdot \psi_i'(w_i;\varepsilon) \geq 0, \tag{44}$$

$$i \in S_\varepsilon^-(\mathbf{w}) \iff \psi_i(w_i;\varepsilon) \leq 0 \text{ and } -\alpha\psi_i(w_i;\varepsilon) \geq 0 > -[\nabla \ell(\mathbf{w})]_i \cdot \psi_i'(w_i;\varepsilon) \tag{45}$$

We apply the descent lemma on the M_ℓ-smooth function ℓ and obtain

$$\begin{aligned}
\ell(\mathbf{w}^{(k+1)}) \leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma_k}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma_k \sum_{i \in S_\varepsilon^+(\mathbf{w}^{(k)})} u_i^{(k)} [\nabla \ell(\mathbf{w}^{(k)})]_i \\
+ \gamma_k \sum_{i \in S_\varepsilon^-(\mathbf{w}^{(k)})} u_i^{(k)} [\nabla \ell(\mathbf{w}^{(k)})]_i + \frac{\gamma_k^2 M_\ell}{2} \sum_{i \in S_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2 \qquad (46)\\
\leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma_k}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma_k \sum_{i \in S_\epsilon^+(\mathbf{w}^{(k)})} |u_i^{(k)}|^2 \\
- \gamma_k \sum_{i \in S_\epsilon^-(\mathbf{w}^{(k)})} |u_i^{(k)}| |[\nabla \ell(\mathbf{w}^{(k)})]_i| + \frac{\gamma_k^2 M_\ell}{2} \sum_{i \in S_\varepsilon(\mathbf{w}^{(k)})} M_c^2 \qquad (47)\\
\leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma_k}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma_k \sum_{i \in S_\epsilon^+(\mathbf{w}^{(k)})} |u_i^{(k)}|^2 + \frac{\gamma_k^2 M_\ell M_c^2 d}{2}. \qquad (48)
\end{aligned}$$

Completing the entire update direction with the skewing force, we bound $\|\mathbf{v}^{(k)}\|^2$ by

$$\begin{aligned}
\frac{\gamma_k}{2}\|\mathbf{v}^{(k)}\|^2 = \frac{\gamma_k}{2} \sum_{i \notin S_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \frac{\gamma_k}{2} \sum_{i \in S_\varepsilon(\mathbf{w}^{(k)})} |u_i^{(k)}|^2 \qquad (49)\\
\leq \ell(\mathbf{w}^{(k)}) - \ell(\mathbf{w}^{(k+1)}) + \frac{3\gamma_k}{2} \sum_{i \in S_\epsilon(\mathbf{w}^{(k)})} |u_i^{(k)}|^2 + \frac{\gamma_k^2 M_\ell M_c^2 d}{2} \qquad (50)\\
\leq \ell(\mathbf{w}^{(k)}) - \ell(\mathbf{w}^{(k+1)}) + \frac{3\gamma_k d M_c^2}{2} + \frac{\gamma_k^2 M_\ell M_c^2 d}{2}. \qquad (51)
\end{aligned}$$

Summing from epoch k_0 to T, we have

$$\sum_{k=k_0}^{T} \frac{\gamma_k}{2}\|\mathbf{v}^{(k)}\|^2 \leq \ell(\mathbf{w}^{(k_0)}) - \ell(\mathbf{w}^{(K+1)}) + \frac{3dM_c^2}{2} \sum_{k=k_0}^{T} \gamma_k + \frac{M_\ell M_c^2 d}{2} \sum_{k=k_0}^{T} \gamma_k^2 \tag{52}$$

$$\frac{1}{\sum_{k=k_0}^{T} \gamma_k} \sum_{k=k_0}^{T} \frac{\gamma_k}{2}\|\mathbf{v}^{(k)}\|^2 \leq \frac{\ell(\mathbf{w}^{(k_0)}) - \ell(\mathbf{w}^{(T+1)})}{\sum_{k=k_0}^{T} \gamma_k} + \frac{3dM_c^2}{2} + \frac{M_\ell M_c^2 d}{2} \frac{\sum_{k=k_0}^{T} \gamma_k^2}{\sum_{k=k_0}^{T} \gamma_k}. \tag{53}$$

As T tends to infinity, we have

$$\frac{1}{\sum_{k=k_0}^{T} \gamma_k} \sum_{k=k_0}^{T} \frac{\gamma_k}{2}\|\mathbf{v}^{(k)}\|^2 \longrightarrow \mathcal{O}(M_c^2 d). \tag{54}$$

□

E PROOF OF THEOREM 5 AND 6

Definition 5 (Categorization of the Update Direction). *Consider the update in each iteration k, $\widehat{\mathbf{v}}^{(k)}$, we categorize the update direction $\widehat{\mathbf{v}}^{(k)}$ into four types, with the indices collected by $\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)})$, $\hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})$, $\hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)})$, $\hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)})$ and $\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)}) \cup \hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)}) \cup \hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)}) \cup \hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)}) = [d]$. Formally, for the update on i-th coordinate*

$$i \in \hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)}) \Longleftrightarrow \widehat{v}_i^{(k)} = -[\widehat{\nabla}\ell(\mathbf{w}^{(k)})]_i \text{ and } v_i^{(k)} = -[\nabla\ell(\mathbf{w}^{(k)})]_i. \tag{55}$$
$$i \in \hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)}) \Longleftrightarrow \widehat{v}_i^{(k)} = -[\widehat{\nabla}\ell(\mathbf{w}^{(k)})]_i \text{ and } \mathbf{v}_i^{(k)} = u_i^{(k)}. \tag{56}$$
$$i \in \hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)}) \Longleftrightarrow \widehat{v}_i^{(k)} = \widehat{u}_i^{(k)} = v_i^{(k)} = u_i^{k}. \tag{57}$$
$$i \in \hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)}) \Longleftrightarrow \widehat{v}_i^{(k)} = \widehat{u}_i^{(k)} \text{ and } v_i^{(k)} = -[\nabla\ell(\mathbf{w}^{(k)})]_i. \tag{58}$$

Theorem 5 (Asymptotic Convergence with Lipschitz Continuity and Robbins-Monro Stepsizes). *Assuming that $0 < \varepsilon \le \inf_{1\le i\le d} \inf_{1\le j<K_i} |q_i^{(j)} - q_i^{(j+1)}|^2/4$ and A1, A2, A3, A5 holds, where $\{q_i^{(j)}\}$ are the quantization levels, then $\lim_{K\to\infty} \mathrm{d}(\mathbf{w}^{(\tau_K)}, \mathcal{Z}_\varepsilon) = 0$ almost surely, where τ_K is a randomized index at iteration K taking value from k_0 to K, with each value generated with the probability proportional to the stepsizes $(\gamma_k)_{k_0\le k\le K}$.*

Proof. Fix any $\delta > 0$. By Lemma 2, there exists k_1 such that $\sup_{k\ge k_0} \mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) \le \delta$ and thus $\mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) \le \delta$ for all $k \ge k_0$. By assumption A3, there also exists k_2 such that $\gamma_k \le 1/M_\ell$. Setting $k_0 := \max\{k_1, k_2\}$ should satisfy the required conditions. By Lemma 4, we have $|u_i^{(k)}| \le 2\alpha\delta$ for all $k \ge k_0$.

Now, we focus on the convergence argument. First, apply the descent lemma on the M_ℓ-smooth function ℓ with care. We have

$$\mathbb{E}_{\mathbf{w}^{(k)}}[\ell(\mathbf{w}^{(k+1)})] \le \ell(\mathbf{w}^{(k)}) + \gamma_k \mathbb{E}_{\mathbf{w}^{(k)}}[(\widehat{\mathbf{v}}^{(k)})^\top]\nabla\ell(\mathbf{w}^{(k)}) + \frac{\gamma_k^2 M_\ell}{2}\mathbb{E}_{\mathbf{w}^{(k)}}[\|\widehat{\mathbf{v}}^{(k)}\|^2] \tag{59}$$

$$\begin{aligned}
&= \ell(\mathbf{w}^{(k)}) - \gamma_k \sum_{i\in\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2 + \frac{\gamma_k^2 M_\ell}{2}\mathbb{E}_{\mathbf{w}^{(k)}}\left[\sum_{i\in\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\widehat{\nabla}\ell(\mathbf{w}^{(k)})]_i^2\right] \\
&\quad + \gamma_k \sum_{i\in\hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)})} u_i^{(k)}[\nabla\ell(\mathbf{w}^{(k)})]_i + \frac{\gamma_k^2 M_\ell}{2} \sum_{i\in\hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)})} (u_i^{(k)})^2 && (60)\\
&\le \ell(\mathbf{w}^{(k)}) - \gamma_k \sum_{i\in\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2 + \frac{\gamma_k^2 M_\ell}{2}\left(\sum_{i\in\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2 + \sigma^2\right) \\
&\quad + \gamma_k \sum_{i\in\hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)})} 2\alpha\delta L_\ell + \frac{\gamma_k^2 M_\ell}{2} \sum_{i\in\hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)})} 4\alpha^2\delta^2 && (61)\\
&\le \ell(\mathbf{w}^{(k)}) - \frac{\gamma_k}{2} \sum_{i\in\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2 + \frac{M_\ell\gamma_k^2\sigma^2}{2} + 2dL_\ell\gamma_k\alpha\delta + 2dM_\ell\gamma_k^2\alpha^2\delta^2, && (62)
\end{aligned}$$

where we have applied $1 - \gamma_k M_\ell/2 > 1/2$ for all $k \ge k_0$ in the last step.

Now, we can take full expectation with constant $\mathtt{a} \ge 2dL_\ell\alpha + 2dM_\ell\gamma_k\alpha^2\delta$ and derive

$$\frac{\gamma_k}{2}\mathbb{E}\left[\sum_{i\in\hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)})\cup\hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\nabla\ell(\mathbf{w}^{(k)})]_i^2\right] \le \mathbb{E}[\ell(\mathbf{w}^{(k)})] - \mathbb{E}[\ell(\mathbf{w}^{(k+1)})] + \frac{M_\ell\gamma_k^2\sigma^2}{2} + 2dL_\ell\gamma_k\alpha\delta + 2dM_\ell\gamma_k^2\alpha^2\delta^2 \tag{63}$$

$$\le \mathbb{E}[\ell(\mathbf{w}^{(k)})] - \mathbb{E}[\ell(\mathbf{w}^{(k+1)})] + \frac{M_\ell\gamma_k^2\sigma^2}{2} + \mathtt{a}\gamma_k\delta. \tag{64}$$

Summing from epoch k_0 to K, we have

$$\sum_{k=k_0}^{K} \frac{\gamma_k}{2} \mathbb{E}\left[\sum_{i \in \hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)}) \cup \hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 \right] \leq \left(\mathbb{E}[\ell(\mathbf{w}^{(k_0)})] - \mathbb{E}[\ell(\mathbf{w}^{(K+1)})] \right) + \frac{M_\ell \sigma^2}{2} \sum_{k=k_0}^{K} \gamma_k^2 + \mathtt{a}\delta \sum_{k=k_0}^{K} \gamma_k. \tag{65}$$

Let $H_K = \sum_{k=k_0}^{K} \gamma_k$ and hence $\tau_K = k$ with probability γ_k / H_K, we can take the expectation of the randomized squared-gradient:

$$\mathbb{E}\left[[\nabla \ell(\mathbf{w}^{(\tau_K)})]_i^2 \right] = \sum_{k=k_0}^{K} \mathbb{P}(\tau_K = k) \mathbb{E}\left[[\nabla \ell(\mathbf{w}^{(k)})]_i^2 \right] = \sum_{k=k_0}^{K} \frac{\gamma_k \mathbb{E}\left[[\nabla \ell(\mathbf{w}^{(k)})]_i^2 \right]}{H_K} = \frac{2}{H_K} \sum_{k=k_0}^{K} \frac{\gamma_k}{2} \mathbb{E}\left[[\nabla \ell(\mathbf{w}^{(k)})]_i^2 \right]. \tag{66}$$

$$\begin{aligned}
\mathbb{E}\left[\sum_{i \in \hat{S}_{1,\varepsilon}(\mathbf{w}^{(\tau_K)}) \cup \hat{S}_{2,\varepsilon}(\mathbf{w}^{(\tau_K)})} [\nabla \ell(\mathbf{w}^{(\tau_K)})]_i^2 \right] &= \frac{2}{H_K} \sum_{k=k_0}^{K} \frac{\gamma_k}{2} \mathbb{E}\left[\sum_{i \in \hat{S}_{1,\varepsilon}(\mathbf{w}^{(k)}) \cup \hat{S}_{2,\varepsilon}(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 \right] && (67) \\
&\leq \frac{2}{H_K} \left(\mathbb{E}[\ell(\mathbf{w}^{(k_0)})] - \mathbb{E}[\ell(\mathbf{w}^{(K+1)})] \right) \\
&\quad + \frac{L_\ell \sigma^2}{H_K} \sum_{k=k_0}^{K} \gamma_k^2 + \frac{2\mathtt{a}\delta}{H_K} \sum_{k=k_0}^{K} \gamma_k && (68) \\
&\xrightarrow{K \to \infty} 2\mathtt{a}\delta, && (69)
\end{aligned}$$

since A3 ensures that $H_k \to \infty$ as $K \to \infty$.

We then consider the case where $i \in \hat{S}_{3,\varepsilon}(\mathbf{w}^{(k)})$. This case is resolved by noting that the true update direction is $v_i^{(k)} = u_i^{(k)} \leq 2\alpha\delta$, given that $k \geq k_0$.

Lastly, we consider the case where $i \in \hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)})$. For this case to happen, we must first have $\psi_i(w_i^{(k)}; \varepsilon) \leq 0$. Let $c_- < c_+$ be such that $\psi_i(c_-; \varepsilon) = \psi_i(c_+; \varepsilon) = 0$. By Lemma 2, for all k large enough, there are three possible cases: (a) $w_i^{(k)}$ stays at the left of $[c_-, c_+]$ or $w_i^{(k)}$ stays at the right of $[c_-, c_+]$, or (b) $w_i^{(k)}$ visits $[c_-, c_+]$ infinitely often.

If there is no k_1 such that $w_i^{(k)} \in (c_-, c_+)$ for all $k \geq k_1$, then there exists a subsequence $\{w_i^{(j)}\}_{j \geq 1}$ such that $w_i^{(j)} \to c_-$ or $w_i^{(j)} \to c_+$, since $\limsup_k w_i^{(k)} \leq c_+$, $\liminf_k w_i^{(k)} \geq c_-$. As a result, either $w_i^\infty \to c_-$ or $w_i^\infty \to c_+$, as $\mathrm{d}(\mathbf{w}^{(k)}, C_\varepsilon) \to 0$ (if w_i ever converges). Otherwise, $\hat{S}_{4,\varepsilon}(\mathbf{w}^{(k)}) = \emptyset$ for every $k \geq k_1$.

We verify the stationarity condition for w_i^∞. It suffices to consider the case where $w_i^\infty = c_-$ without loss of generality. Violation of the stationary condition only happens when $-[\nabla \ell(\mathbf{w}^\infty)]_i \cdot \psi_i'(c_-; \varepsilon) > 0$. However, the algorithm will take an update with $-[\nabla \ell(\mathbf{w}^\infty)]_i \neq 0$, which violates $w_i^{(k)}$'s convergence to c_-.

Denote $\bar{\bar{\mathbf{w}}}$ as a result of projecting $\mathbf{w}$ to C_ε for coordinates $i \in S_{4,\varepsilon}(\mathbf{w})$. It is clear that the smoothness condition gives $\|\nabla \ell(\mathbf{w}) - \nabla \ell(\bar{\bar{\mathbf{w}}})\| \leq M_\ell \|\bar{\bar{\mathbf{w}}} - \mathbf{w}\| \leq M_\ell \delta$. This implies $\|\nabla \ell(\mathbf{w})\| \leq \|\nabla \ell(\bar{\bar{\mathbf{w}}})\| + M_\ell \delta$ and hence $\|\nabla \ell(\mathbf{w})\|^2 \leq \|\nabla \ell(\bar{\bar{\mathbf{w}}})\|^2 + 2M_\ell \delta \|\nabla \ell(\bar{\bar{\mathbf{w}}})\| + M_\ell^2 \delta^2$.

$$\begin{aligned}
\mathbb{E}\left[\|\nabla \ell(\mathbf{w}^{(\tau_K)})\|^2 \right] &= \mathbb{E}\left[\sum_{i \in [n]} [\nabla \ell(\mathbf{w}^{(\tau_K)})]_i^2 \right] && (70) \\
&= \mathbb{E}\left[\sum_{i \in \hat{S}_{1,\varepsilon}(\mathbf{w}^{(\tau_K)}) \cup \hat{S}_{2,\varepsilon}(\mathbf{w}^{(\tau_K)})} [\nabla \ell(\mathbf{w}^{(\tau_K)})]_i^2 \right] \\
&\quad + \mathbb{E}\left[\sum_{i \in \hat{S}_{3,\varepsilon}(\mathbf{w}^{(\tau_K)})} [\nabla \ell(\mathbf{w}^{(\tau_K)})]_i^2 \right] + \mathbb{E}\left[\sum_{i \in \hat{S}_{4,\varepsilon}(\mathbf{w}^{(\tau_K)})} [\nabla \ell(\mathbf{w}^{(\tau_K)})]_i^2 \right] && (71)
\end{aligned}$$

$$\leq 2\mathtt{a}\delta + (2\alpha\delta)^2 + M_\ell^2\delta^2 + 2M_\ell\delta\mathbb{E}\left[\sum_{i\in\hat{S}_{4,\varepsilon}(\mathbf{w}^{(\tau_K)})}\left|\left[\nabla\ell(\bar{\bar{\mathbf{w}}}^{(\tau_K)})\right]_i\right|\right]$$

$$+\mathbb{E}\left[\sum_{i\in\hat{S}_{4,\varepsilon}(\mathbf{w}^{(\tau_K)})}\left[\nabla\ell(\bar{\bar{\mathbf{w}}}^{(\tau_K)})\right]_i^2\right] \quad \text{(as } K \text{ approaches infinity)} \tag{72}$$

$$\xrightarrow{K\to\infty\Rightarrow\delta\to 0}\mathbb{E}\left[\sum_{i\in\hat{S}_{4,\varepsilon}(\mathbf{w}^{(\tau_K)})}\left[\nabla\ell(\bar{\bar{\mathbf{w}}}^{(\tau_K)})\right]_i^2\right]. \tag{73}$$

From the above we know that we can be arbitrarily close to a stationary point in $\mathcal{Z}_\varepsilon$. □

Theorem 6 (Asymptotic Convergence with Lipschitz Smoothness and Robbins-Monro Stepsizes). *Assuming that $0 < \varepsilon \leq \inf_{1\leq i\leq d}\inf_{1\leq j<K_i}|q_i^{(j)} - q_i^{(j+1)}|^2/4$, and A1, A3, A5, A6 holds, where $\{q_i^{(j)}\}$ are the quantization levels, then $\lim_{k\to\infty}\mathrm{d}(\mathbf{w}^{(k)},\mathcal{Z}_\varepsilon) = 0$ almost surely.*

Proof. The `ASkewSGD` update is

$$\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} + \gamma_k\widehat{\mathbf{v}}^{(k)}, \tag{74}$$

where $\widehat{\mathbf{v}}^{(k)} = \mathbf{s}_{\varepsilon,\alpha}(\widehat{\nabla\ell}(\mathbf{w}^{(k)}),\mathbf{w}^{(k)})$, with the mean direction is $\mathbf{h}(\mathbf{w}) = \mathbb{E}[\mathbf{s}_{\varepsilon,\alpha}(\widehat{\nabla\ell}(\mathbf{w}),\mathbf{w})|\mathbf{w}]$.

Consider the Stochastic Approximation (SA) Scheme, where we express the update as:

$$\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} + \gamma_k(\mathbf{h}(\mathbf{w}^{(k)}) + \boldsymbol{\xi}^{(k)}), \tag{75}$$

where $\boldsymbol{\xi}^{(k)} = \widehat{\mathbf{v}}^{(k)} - \mathbf{h}(\mathbf{w}^{(k)})$ is a martingale difference noise.

Define the Lyapunov function

$$\mathcal{L}(\mathbf{w}) = \ell(\mathbf{w}) + \frac{\alpha}{2}\sum_{i=1}^{d}[\min(0,\psi_i(\mathbf{w}))]^2. \tag{76}$$

Since $\ell \in C^1$, and the penalty term $[\min(0,\psi_i(w_i;\varepsilon))]^2$ is C^1 due to the piecewise C^1 structure of ψ_i and the derivative continuity at boundaries, we know that $\mathcal{L}$ is C^1. Note that ℓ is also radially bounded (if ℓ is also coercive), with the gradient

$$\nabla\mathcal{L}(\mathbf{w}) = \nabla\ell(\mathbf{w}) + \mathbf{r}(\mathbf{w}), \text{where } r_i(\mathbf{w}) = \begin{cases} 0, & \psi_i(w_i;\varepsilon) > 0, \\ \alpha\psi_i(w_i;\varepsilon)\psi_i'(w_i;\varepsilon), & \psi_i(w_i;\varepsilon) \leq 0. \end{cases} \tag{77}$$

Moreover, $\mathbf{h}(\mathbf{w}) = \mathbf{0}$ if and only if $\mathbf{w} \in \mathcal{Z}_\varepsilon$ (direct from Lemma 3).

We need to prove the following claim.

Claim. Almost surely, $\langle\nabla\mathcal{L}(\mathbf{w}),\mathbf{h}(\mathbf{w})\rangle \leq -\mathtt{b}\|\mathbf{h}(\mathbf{w})\|^2$ for some $\mathtt{b} > 0$ when $\mathbf{h}(\mathbf{w}) \neq 0$.

Case 1. If w_i is strictly feasible, i.e. $\psi_i(\mathbf{w}) > 0$, then $[\nabla\mathcal{L}(\mathbf{w})]_i = [\nabla\ell(\mathbf{w})]_i$ and $[h(\mathbf{w})]_i = -[\nabla\ell(\mathbf{w})]_i$. Hence,

$$[\nabla\mathcal{L}(\mathbf{w})]_i[\mathbf{h}(\mathbf{w})]_i = -[\nabla\ell(\mathbf{w})]_i^2 = -[\mathbf{h}(\mathbf{w})]_i^2. \tag{78}$$

Case 2. If w_i is not strictly feasible, i.e. $\psi_i(\mathbf{w}) \leq 0$, then $[\nabla\mathcal{L}(\mathbf{w})]_i = [\nabla\ell(\mathbf{w})]_i + \alpha\psi_i(w_i;\varepsilon)\psi_i'(w_i;\varepsilon)$.

(a) The gradient condition holds, i.e. $-[\nabla\ell(\mathbf{w})]_i\cdot\psi_i'(w_i;\varepsilon) \geq -\alpha\psi_i(w_i;\varepsilon) \geq 0$. Then, $[\mathbf{h}(\mathbf{w})]_i = -[\nabla\ell(\mathbf{w})]_i$, and

$$[\nabla\mathcal{L}(\mathbf{w})]_i\cdot[\mathbf{h}(\mathbf{w})]_i = -[\nabla\ell(\mathbf{w})]_i^2 - \alpha\psi_i(w_i;\varepsilon)\psi_i'(w_i;\varepsilon)\cdot[\nabla\ell(\mathbf{w})]_i \tag{79}$$

$$\leq -[\nabla\ell(\mathbf{w})]_i^2 - \alpha^2\psi_i^2(w_i;\varepsilon) \leq -[\mathbf{h}(\mathbf{w})]_i^2, \tag{80}$$

where

$$-\alpha\psi_i(w_i;\varepsilon)\psi_i'(w_i;\varepsilon)\cdot[\nabla\ell(\mathbf{w})]_i \leq -\alpha\psi_i(w_i;\varepsilon)\cdot(\alpha\psi_i(w_i;\varepsilon)) = -\alpha^2\psi_i(w_i;\varepsilon)^2. \tag{81}$$

(b) The gradient condition fails, i.e. $-[\nabla \ell(\mathbf{w})]_i \cdot \psi_i'(w_i;\varepsilon) < -\alpha\psi_i(w_i;\varepsilon)$. Then, $[\mathbf{h}(\mathbf{w})]_i = \mathrm{clip}(-\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon), M_c)$.
If $|[\mathbf{h}(\mathbf{w})]_i| \le M_c$, then

$$
\begin{aligned}
[\nabla \mathcal{L}(\mathbf{w})]_i \cdot [\mathbf{h}(\mathbf{w})]_i &= ([\nabla \ell(\mathbf{w})]_i + \alpha\psi_i(w_i;\varepsilon)\psi_i'(w_i;\varepsilon)) \cdot (-\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon)) && (82)\\
&= -\alpha \frac{\psi_i(w_i;\varepsilon)\cdot[\nabla \ell(\mathbf{w})]_i}{\psi_i'(w_i;\varepsilon)} - \alpha^2\psi_i(w_i;\varepsilon) && (83)\\
&< -\alpha^2\psi_i^2(w_i;\varepsilon)\left(\frac{1}{(\psi_i'(w_i;\varepsilon))^2} + 1\right) && (84)\\
&\le -\left(\frac{\alpha\psi_i(w_i;\varepsilon)}{\psi_i'(w_i;\varepsilon)}\right)^2 && (85)\\
&= -[\mathbf{h}(\mathbf{w})]_i^2. && (86)
\end{aligned}
$$

Otherwise, assume that clipping is activated and $[\mathbf{h}(\mathbf{w})]_i = \varsigma M_c$, where $\varsigma = \mathrm{sign}(-\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon))$. Since the case where $\psi_i'(w_i;\varepsilon) = 0$ is measure-zero, then almost surely,

$$
\begin{aligned}
[\nabla \mathcal{L}(\mathbf{w})]_i \cdot [\mathbf{h}(\mathbf{w})]_i &= [\nabla \mathcal{L}(\mathbf{w})]_i \cdot (-\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon)) \cdot \frac{\varsigma M_c}{-\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon)} && (87)\\
&\le -(\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon))^2 \cdot \frac{\varsigma M_c}{-\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon)} && (88)\\
&= (\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon)) \cdot \varsigma M_c && (89)\\
&\le -\varsigma^2 M_c^2 \quad (\text{since } |\alpha\psi_i(w_i;\varepsilon)/\psi_i'(w_i;\varepsilon)| \ge M_c) && (90)\\
&= -M_c^2. && (91)
\end{aligned}
$$

Thus, $\mathcal{L}(\mathbf{w}(t))$ is almost surely non-increasing along trajectories. Moreover, $\mathcal{L}$ is radially unbounded (due to A6), then $\mathcal{L}(\mathbf{w}(t))$ converges to some $\mathcal{L}^* \ge \inf \mathcal{L}$ as $t \to \infty$.

Define the equilibrium set:

$$
= \left\{\mathbf{w} : \frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L}(\mathbf{w}) = 0\right\} \overset{\text{a.s.}}{=} \{\mathbf{w} : h(\mathbf{w}) = 0\} = \mathcal{Z}_\varepsilon
$$

By LaSalle's Invariance Principle, the largest invariant set in E is $\mathcal{Z}_\varepsilon$ (since $h(\mathbf{w}) = 0$ implies $\mathbf{w}(t)$ is constant). Hence,

$$
\lim_{t\to\infty} \mathrm{d}(\mathbf{w}(t), \mathcal{Z}_\varepsilon) = 0 \quad \text{a.s.}. \tag{92}
$$

For any $\epsilon > 0$, choose $\xi > 0$ such that

$$
\mathbf{w}(0) \in B_\xi(\mathcal{Z}_\varepsilon) \implies \mathcal{L}(\mathbf{w}(0)) < \min_{\mathbf{w}\in\partial B_\epsilon(\mathcal{Z}_\varepsilon)} \mathcal{L}(\mathbf{w}). \tag{93}
$$

Since $\mathcal{L}$ decreases along trajectories, $\mathbf{w}(t) \in B_\epsilon(\mathcal{Z}_\varepsilon)$ for all $t \ge 0$, and the set $\mathcal{Z}_\varepsilon$ is almost surely globally asymptotically stable.

Via Kushner-Clark theorem (Fort & Gilles Pagès, 1996), we verify that $\{\mathbf{w}^{(k)}\}$ is bounded a.s., ℓ is smooth by A1, $\mathbb{E}[\boldsymbol{\xi}^{(k)}|\mathcal{F}_k] = 0$, $\mathbb{E}[\|\boldsymbol{\xi}^{(k)}\|^2|\mathcal{F}_k] \le \sigma^2$ by A5, $\sum \gamma_k = \infty, \sum \gamma_k^2 < \infty$ by A3, and $\mathcal{Z}_\varepsilon$ is almost surely globally asymptotically stable for ODE, and conclude that

$$
\lim_{k\to\infty} \mathrm{d}(\mathbf{w}^{(k)}, \mathcal{Z}_\varepsilon) = 0 \quad \text{a.s.}. \tag{94}
$$

□

F EXPERIMENTAL SETUP AND RESULTS

For `BinaryConnect` (Courbariaux et al., 2015), we adopt the deterministic version as below:

Algorithm 2: `BinaryConnect` for BNN Training

for $k = 0, 1, \ldots$ **do**
- Obtain the binarized weights $\mathbf{w}_{\mathrm{bin}}^{(k)} = \mathrm{sign}(\mathbf{w}^{(k)})$.
- Obtain a stochastic gradient $\widehat{\nabla}\ell(\mathbf{w}_{\mathrm{bin}}^{(k)}) = 1/N_b \sum_{i=1}^{N_b} \nabla \ell_{j_i}(\mathbf{w}_{\mathrm{bin}}^{(k)})$.
- $\mathbf{w}^{(k+1)} \leftarrow \mathrm{clamp}(\mathbf{w}^{(k)} - \gamma_k \widehat{\nabla}\ell(\mathbf{w}_{\mathrm{bin}}^{(k)}), -1, +1)$.

We also extend `BinaryConnect` to the k-bit ($k > 1$) case, whose implementation is shown as below:

Algorithm 3: `MultiBitConnect` for QNN Training

for $k = 0, 1, \ldots, T$ **do**
- Obtain the quantized weights $\mathbf{w}_{\mathrm{qt}}^{(k)} = \mathrm{clamp}(\mathrm{round}(\mathbf{w}^{(k)}), -2^{k-1}, 2^{k-1} - 1)$.
- Obtain a stochastic gradient with quantized weights $\widehat{\nabla}\ell(\mathbf{w}_{\mathrm{qt}}^{(k)}) = 1/N_b \sum_{i=1}^{N_b} \nabla \ell_{j_i}(\mathbf{w}_{\mathrm{qt}}^{(k)})$.
- Update the full-precision weights: $\mathbf{w}^{(k+1)} \leftarrow \mathrm{clamp}(\mathbf{w}^{(k)} - \gamma_k \widehat{\nabla}\ell(\mathbf{w}_{\mathrm{qt}}^{(k)}), -2^{k-1}, 2^{k-1} - 1)$.

For `ProxQuant`, we use the version with L_2 proximal operator, i.e.

$$\mathrm{prox}_R(\mathbf{w}) := \underset{\widetilde{\mathbf{w}} \in \mathbb{R}^d}{\arg\min} \left\{ (1/2)\|\widetilde{\mathbf{w}} - \mathbf{w}\|^2 + \lambda \inf_{\widehat{\mathbf{w}} \in \mathcal{Q}} \|\widetilde{\mathbf{w}} - \widehat{\mathbf{w}}\|^2 \right\}. \tag{95}$$

This formulation admits an explicit solution, which is

$$\widetilde{\mathbf{w}} := \frac{\mathbf{w} + 2\lambda \cdot \mathrm{sign}(\mathbf{w})}{1 + 2\lambda}. \tag{96}$$

We can directly extend this to the general k-bit case, which gives:

Algorithm 4: `ProxQuant` for QNN Training

for $k = 0, 1, \ldots, T$ **do**
- Obtain the regularization strength at iteration k: $\lambda_k = \max\{0, \lambda(k - t_{PQ})/(T - t_{PQ})\}$
- Obtain a stochastic gradient $\widehat{\nabla}\ell(\mathbf{w}^{(k)}) = 1/N_b \sum_{i=1}^{N_b} \nabla \ell_{j_i}(\mathbf{w}^{(k)})$.
- Obtain the closest quantization level $\mathbf{q}^{(k)}$ for each parameter $\mathbf{w}^{(k)}$.
- Update the full-precision weights: $\mathbf{w}^{(k+1)} \leftarrow (\mathbf{w}^{(k)} + \lambda_k \cdot \mathbf{q}^{(k)})/(1 + \lambda_k)$.
- Perform clipping for each parameter that ensures each w_i does not exceed its quantization range.

F.1 Supplementary Results of Two Moons Classification

Table 4: Performance after 60 epochs (average and standard deviation over 15 random experiments).

Method	**Test Loss**	**Quantized Test Loss**	**Quantized Accuracy**
Full Precision [W32/A32]	0.02 ± 0.0413	5.44 ± 2.6426	$49.85\% \pm 5.1620$
Deterministic `BinaryConnect` [W1/A32]	0.80 ± 0.4973	$\mathbf{0.03 \pm 0.0731}$	$98.77\% \pm 4.0333$
`ProxQuant` [W1/A32]	$\mathbf{0.01 \pm 0.0058}$	0.28 ± 0.5932	$96.03\% \pm 6.8739$
`ASkewSGD` [W1/A32]	0.01 ± 0.0111	0.05 ± 0.0719	$\mathbf{98.80\% \pm 2.3833}$

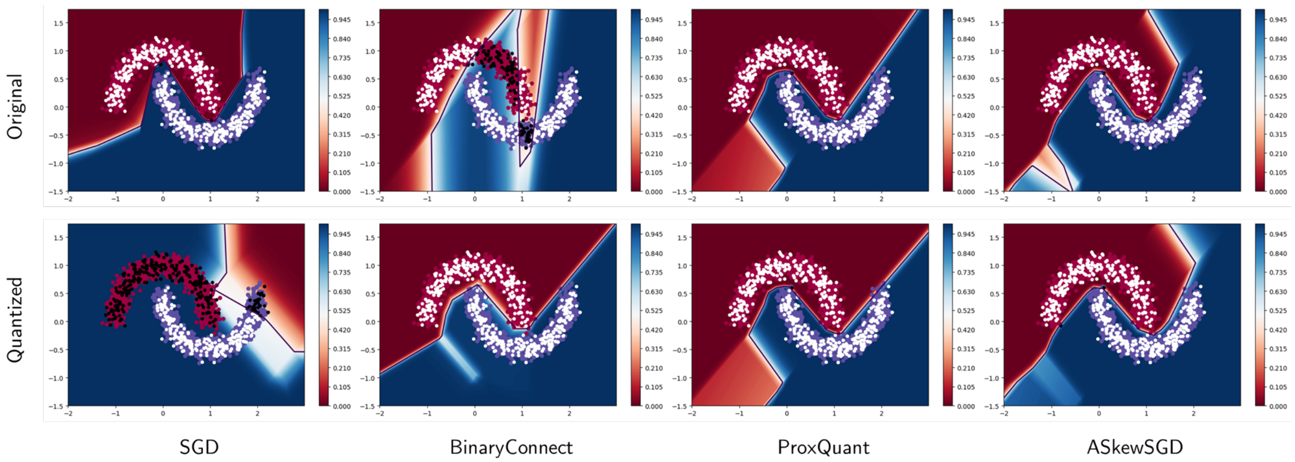

Figure 7: The representative results of the 2 moons classification problem.

The original net has its contour completely perturbed by the weight quantization process. `BinaryConnect` is aware of the quantization scheme, thus generalizing well on the test set, but the original model is completely off from the groundtruth. `ASkewSGD` and `ProxQuant`, on the other hand, have only a slight skew away from the sensible region, and the quantized model is still able to generalize well on the test set, which demonstrates the strength of the two optimization methods. However, we observe from Table 4 that `ProxQuant` can still perform inconsistently even on this simple task.

F.2 Supplementary Results of MNIST

Table 5: Performance on MNIST after 50 epochs over 5 random experiments.

Method	Test Loss	Test Accuracy	Quantized Test Accuracy	Top-1 (Quantized)
Full Precision [W32/A32]	**0.065 ± 0.001**	**$98.01\% \pm 0.002$**	N/A	**98.20%***
`BinaryConnect` [W1/A32]	2.289 ± 0.001	$58.70\% \pm 4.058$	$93.26\% \pm 0.455$	93.66%
`ProxQuant` [W1/A32]	0.089 ± 0.001	$97.25\% \pm 0.034$	$93.75\% \pm 1.100$	94.94%
`ASkewSGD` [W1/A32]	143.501 ± 6.343	$95.78\% \pm 0.298$	**$94.47\% \pm 0.508$**	**95.14%**
`BinaryConnect` [W2/A32]	2.288 ± 0.002	$86.67\% \pm 0.855$	$94.25\% \pm 0.485$	94.80%
`ProxQuant` [W2/A32]	0.144 ± 0.003	$95.74\% \pm 0.068$	$87.48\% \pm 6.237$	93.33%
`ASkewSGD` [W2/A32]	0.068 ± 0.001	$97.94\% \pm 0.075$	**$97.26\% \pm 0.252$**	**97.55%**

Note that we do not quantize the full precision net during its evaluation.